

TARVIA Q&A Benchmark

Reviewer-grade reasoning questions on the RelA/RelB — $\alpha V\beta 3$ axis in HGSOC

75 records · 30 mechanistic · 30 data interpretation · 15 hypothesis

Source: Lujano-Olazaba et al., Cell Reports 2026

Point-weighted scoring rubric, LifeSciBench-style

Scoring Rubric Methodology

Each question's four required reasoning elements are scored as independent, point-weighted criteria rather than a single pass/fail judgment — the format used in the LifeSciBench benchmark (OpenAI & Tacit Labs, 2026) for grading open-ended expert-level tasks. A response earns full credit only by independently satisfying each bracketed criterion; partial coverage yields partial, auditable credit.

Rubric-writing principles applied

- Specificity — each criterion describes one concrete, checkable claim.
- Atomicity — no criterion bundles more than one judgment.
- Evaluability — satisfied or not, directly from the response text.
- Grounding — every criterion traces to the manuscript's reported data or the reviewer exchange, not outside knowledge.
- Weighting — criteria carrying a causal or falsifiable claim (marked with → in the source key points, or an explicit prediction/test) are weighted higher than restated setup facts; totals run 20 points for medium-difficulty questions and 25 for hard.

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Mechanistic

30 questions · relb-M01–relb-M30

relb-M01 MEDIUM · 20 pts

RelA regulates ITGAV and RelB regulates ITGB3 in HGSOc. Why does this division of labor make surface alphaVbeta3 a coincidence detector, and what does that imply for single-arm targeting?

REFERENCE ANSWER

AlphaVbeta3 is an obligate heterodimer; neither subunit forms a functional receptor alone. If RelA supplies alphaV and RelB supplies beta3, surface receptor requires input from both the classical and alternative NF-kB arms, making the heterodimer a readout of dual-pathway activation rather than either arm alone. This predicts that knocking down one subunit's driver leaves the other subunit intact and available for compensation with other partners (alphaV pairs with beta5, beta1, beta6, beta8), which is precisely why single-arm knockdown reduces but does not eliminate alphaVbeta3, and why RelB knockdown alone failed to improve survival without Cilengitide.

SCORING RUBRIC (20 pts)

- **[+5]** obligate heterodimer requires both subunits
- **[+5]** RelA and RelB each supply one subunit -> dual-arm coincidence detection
- **[+5]** spare alphaV can pair with beta5/beta1 -> compensation
- **[+5]** predicts single-arm knockdown is insufficient; matches shRelB-alone survival failure

Lujano-Olazaba et al., Cell Reports 2026 · Figure 1D, Figure 6B

relb-M02 HARD · 25 pts

RelB is a transcriptional target of RelA. Explain precisely how this confounds the interpretation of shRelA phenotypes, and what pattern of results would still permit a RelA-specific conclusion.

REFERENCE ANSWER

Because RelA drives RELB transcription, shRelA knocks down RelA directly and RelB indirectly, so any shRelA phenotype could be a RelA effect, a RelB effect, or the sum. shRelB is comparatively clean since RelB does not drive RELA. A RelA-specific conclusion survives only where shRelA produces a phenotype that shRelB does not reproduce, or produces the opposite direction. ITGAV downregulation qualifies: it occurs with shRelA but not shRelB, so it cannot be explained by the RelB depletion that shRelA also causes. Conversely, any phenotype shared by both knockdowns is uninterpretable as RelA-specific.

SCORING RUBRIC (25 pts)

- **[+6]** shRelA depletes RelB indirectly; shRelB does not deplete RelA
- **[+7]** shared phenotypes cannot be assigned to RelA
- **[+6]** RelA-specific claims require shRelA-only or opposite-direction effects
- **[+6]** ITGAV meets this bar; the paper lists this confound as a stated limitation

Lujano-Olazaba et al., Cell Reports 2026 · Figure 1D; Limitations

relb-M03 MEDIUM · 20 pts

RelB knockdown downregulates ITGB3 only in 3D culture, not 2D. What does the condition-dependence tell you about the mechanism of RelB-driven ITGB3 transcription?

REFERENCE ANSWER

Condition-dependence means RelB is necessary but not sufficient for ITGB3 transcription; a 3D-restricted co-input is also required. RelB protein is present in 2D, yet its loss does nothing there, so in 2D either RelB is not loaded at the ITGB3 locus or the locus is not permissive. 3D culture (low serum, ultra-low attachment) imposes isolation and nutrient stress, which activates alternative NF- κ B and likely opens the locus or supplies a partner such as p52. The mechanistic claim is therefore stress-gated RelB activity, not constitutive RelB abundance, which fits the paper's framing of RelB as active in quiescent, stressed cells while RelA dominates in proliferative ones.

SCORING RUBRIC (20 pts)

- **[+6]** RelB necessary but not sufficient; requires a 3D/stress-restricted co-input
- **[+4]** argues activity/localization, not abundance, is the regulated variable
- **[+6]** 3D = isolation + nutrient stress $\rightarrow$ alternative NF- κ B activation
- **[+4]** consistent with prior RelA-proliferative / RelB-quiescent division

Lujano-Olazaba et al., Cell Reports 2026 · Figure 1D, Supplemental Figure 1B-C

relb-M04 MEDIUM · 20 pts

TWEAK-Fn14 signaling upregulates α V β 3 but not α V β 5 or α 5 β 1. Given that TWEAK activates alternative NF- κ B, explain why the selectivity is mechanistically expected.

REFERENCE ANSWER

TWEAK-Fn14 signals principally through the alternative arm, stabilizing NIK, driving p100 processing to p52, and mobilizing RelB:p52. If β 3 is the RelB-dependent subunit and β 5 is not, then a purely alternative-pathway stimulus should move α V β 3 and leave α V β 5 untouched. The result is therefore a functional confirmation of the Figure 1 transcriptional assignment using a physiological ligand rather than a genetic knockdown: the selectivity of the input predicts the selectivity of the output. α 5 β 1's non-response has a separate explanation (a surface ceiling), so it does not add independent evidence for pathway selectivity.

SCORING RUBRIC (20 pts)

- **[+5]** TWEAK-Fn14 preferentially activates alternative NF- κ B (NIK/p100 $\rightarrow$ p52/RelB)
- **[+7]** β 3 is RelB-dependent, β 5 is not $\rightarrow$ selective upregulation predicted
- **[+4]** physiological-ligand confirmation of the genetic subunit assignment
- **[+4]** α 5 β 1's non-response is a ceiling artifact, not pathway selectivity evidence

Lujano-Olazaba et al., Cell Reports 2026 · Figure 4B, Supplemental Figure 4C-D

relb-M05 HARD · 25 pts

α V β 3 blockade does not reduce adhesion to fibronectin, yet α V β 3+ cells are dramatically more tumorigenic. Reconcile these two findings mechanistically.

REFERENCE ANSWER

Adhesion and survival signaling are separable integrin functions. α 5 β 1 is the fibronectin adhesion receptor here (blocking it reduces attachment; blocking α V β 3 does not), so α V β 3 is not doing the mechanical anchoring. Its contribution is outside-in survival signaling under stress: unligated or stress-engaged α V β 3 nucleates signaling complexes (FAK/Src, and in other systems a β 3-KRAS-RalB axis) that support anchorage-independence and stress tolerance rather than attachment. That division explains

why alphaVbeta3 is dispensable for adhesion but decisive for surviving isolation stress at 1,500 cells, and it reframes alphaVbeta3 as a stress-tolerance signaling node rather than an adhesion molecule.

SCORING RUBRIC (25 pts)

- **[+7]** adhesion and survival signaling are dissociable integrin functions
- **[+6]** alpha5beta1 performs fibronectin adhesion; alphaVbeta3 does not
- **[+6]** alphaVbeta3 acts via outside-in survival signaling (FAK/Src; beta3-KRAS-RalB precedent)
- **[+6]** explains dispensable-for-adhesion but decisive-for-isolation-stress

Lujano-Olazaba et al., Cell Reports 2026 · Figure 5C-D, Figure 5G-H

relb-M06 MEDIUM · 20 pts

Why does the paper interpret 3D spheroid culture as a stress-tolerance assay rather than simply a stemness assay, and why does that distinction matter for the alphaVbeta3 claim?

REFERENCE ANSWER

3D culture here uses ultra-low attachment plates and low-serum stem cell media, which imposes two specific stressors: loss of matrix anchorage (isolation/anoikis stress) and nutrient limitation. Cells that survive are selected for stress tolerance; stemness marker expression is a correlate, not the defining variable. The distinction matters because alphaVbeta3 is enriched in 3D and further enriched by chemotherapy, two different stressors converging on the same receptor. That convergence is what licenses the claim that alphaVbeta3 marks stress tolerance generally rather than a stemness program specifically, and it is why the same receptor shows up in the low-inoculum in vivo model, which was designed to impose isolation stress.

SCORING RUBRIC (20 pts)

- **[+5]** 3D = ULA + low serum -> isolation/anoikis + nutrient stress
- **[+5]** stemness markers are a correlate of the selection, not the selecting variable
- **[+5]** alphaVbeta3 enriched by two independent stressors (3D and chemotherapy)
- **[+5]** convergence licenses a stress-tolerance claim over a stemness-specific one

Lujano-Olazaba et al., Cell Reports 2026 · Figure 2C, Figure 2E

relb-M07 HARD · 25 pts

RelA and RelB regulate distinct GO programs in 2D but converge on eight shared terms in 3D, including ECM organization. What does condition-dependent convergence imply about NF-kB pathway architecture?

REFERENCE ANSWER

It implies the two arms are not fixed parallel channels but context-gated ones whose outputs overlap under stress. In 2D, RelA runs proliferation-adjacent programs (blood vessel development, migration, matrisome) and RelB runs metabolic ones (hormone levels, lipid localization). In 3D, both converge on ECM organization, cell-cell adhesion, cytokine response and inflammatory signaling. Mechanistically this suggests stress recruits both arms onto a shared matrix-interaction program, likely through shared kappaB elements at ECM loci, while each arm retains private targets (ITGAV vs ITGB3) within that program. The architecture is therefore convergent at the pathway level and divergent at the gene level, which is exactly what makes combination targeting necessary.

SCORING RUBRIC (25 pts)

- **[+5]** arms are context-gated, not fixed parallel channels

- **[+6]** 2D: RelA proliferative/matrisome vs RelB metabolic; 3D: convergence on ECM/adhesion
- **[+6]** likely shared kappaB elements at ECM loci with arm-private genes inside the program
- **[+8]** convergent at pathway level, divergent at gene level -> combination targeting rationale

Lujano-Olazaba et al., Cell Reports 2026 · Figure 1A-C

relb-M08 MEDIUM · 20 pts

Explain the mechanistic logic of using a doxycycline-inducible shRNA rather than constitutive knockdown, and what question this design makes answerable that constitutive knockdown could not.

REFERENCE ANSWER

Constitutive loss of RelA or RelB disrupts tumor initiation outright, so constitutively knocked-down cells would never establish the tumors whose relapse is the object of study; the experiment would collapse into a tumorigenesis assay. Inducible knockdown decouples the two phases: tumors establish with NF-κB intact, then doxycycline switches the factor off at the moment chemotherapy begins. This makes answerable a question constitutive knockdown cannot touch, namely whether RelA or RelB is required for survival and regrowth of an already-established tumor under therapeutic stress, which is the clinically relevant window. It also mirrors the timing of a drug given alongside chemotherapy.

SCORING RUBRIC (20 pts)

- **[+6]** constitutive loss blocks tumor initiation -> no tumor to relapse
- **[+5]** inducible design separates establishment from post-chemotherapy survival
- **[+5]** isolates the requirement in established tumors under therapeutic stress
- **[+4]** temporally mimics a drug dosed with chemotherapy

Lujano-Olazaba et al., Cell Reports 2026 · Figure 6A

relb-M09 HARD · 25 pts

RelA is enriched at the omentum and RelB at the mesentery in relapsed tumors, with roughly 10% RelA+ versus 90% RelB+ cells at the mesentery in vehicle-treated mice. Propose the mechanistic basis for site-specific NF-κB arm usage.

REFERENCE ANSWER

Site-specific arm usage most plausibly reflects site-specific ligand availability rather than cell-intrinsic programming. The omentum is preadipocyte- and macrophage-rich and supplies TNF/IL-1-type classical NF-κB inputs plus growth factors favoring proliferation, matching RelA dominance. The mesentery hosts distinct resident populations, including M2-like LYVE1-high interstitial macrophages that remodel matrix, and TWEAK from myeloid cells drives Fn14-RelB signaling, matching RelB dominance. The prediction that follows is testable and falsifiable: measure TWEAK and TNF protein at each site, and swap the niche by orthotopic transfer. If the same cells flip arm usage with site, the driver is the microenvironment; if not, it is a heritable cell state selected at each site.

SCORING RUBRIC (25 pts)

- **[+5]** site-specific ligand availability rather than cell-intrinsic programming
- **[+6]** omentum: preadipocyte/macrophage TNF-IL1 inputs -> classical/RelA
- **[+7]** mesentery: TWEAK from myeloid cells and LYVE1-hi M2 macrophages -> Fn14/RelB
- **[+7]** falsifiable by measuring site ligands and by niche-swap transfer

relb-M10 MEDIUM · 20 pts

Cilengitide had no effect on cell viability in vitro, alone or with chemotherapy, in either 2D or 3D, yet produced the best in vivo survival when combined with RelB knockdown. What does this dissociation tell you about the drug's mechanism?

REFERENCE ANSWER

The dissociation says Cilengitide is not cytotoxic and does not act cell-autonomously on viability; its benefit requires something present in vivo and absent in culture. Three candidates: an ECM ligand context (vitronectin, fibronectin, COL1A1 supplied by stroma) that must be engaged for blockade to matter; a stromal or immune compartment absent in vitro; or an in vivo-specific magnitude of alphaVbeta3 enrichment, which the paper notes was stronger in vivo than in vitro. The mechanistic implication is that Cilengitide interrupts a tumor-stroma interaction rather than killing cells, which also explains why it only helps in combination and why an in vitro synergy screen would have discarded it.

SCORING RUBRIC (20 pts)

- **[+4]** not cytotoxic; no cell-autonomous viability effect
- **[+6]** requires in vivo ligand/stromal context absent from culture
- **[+4]** alphaVbeta3 enrichment was stronger in vivo than in vitro
- **[+6]** acts by interrupting tumor-stroma interaction -> only works in combination

Lujano-Olazaba et al., Cell Reports 2026 · Supplemental Figure 6B-C, Figure 6B

relb-M11 HARD · 25 pts

Cells growing on decellularized mesentery from carboplatin-treated tumor-naive mice survived carboplatin better than cells on mesentery from vehicle-treated mice. Explain what this design proves and what it deliberately excludes.

REFERENCE ANSWER

The mesentery came from tumor-naive mice, so the matrix change was caused by chemotherapy acting on normal stroma, not by tumor-driven remodeling; and the tissue was decellularized, so the protective effect is carried by matrix composition alone, not by living stromal or immune cells. What it proves is a cell-extrinsic, tumor-independent mechanism: chemotherapy remodels host matrix into a substrate that confers cytotoxic tolerance. What it excludes by design is any contribution from live stroma, secreted factors from living cells, immune interactions, and tumor-induced remodeling. The COL1A1 and vitronectin enrichment in post-treatment human mesentery supplies the candidate composition change, with vitronectin notable as an alphaV-integrin ligand.

SCORING RUBRIC (25 pts)

- **[+7]** tumor-naive donors -> chemo-driven, not tumor-driven, matrix change
- **[+7]** decellularized -> matrix composition alone carries the effect
- **[+5]** proves cell-extrinsic, tumor-independent stromal priming
- **[+6]** excludes live stroma, immune, and secreted-factor contributions; COL1A1/VTN are the candidate ligands

Lujano-Olazaba et al., Cell Reports 2026 · Figure 7H-I

relb-M12 MEDIUM · 20 pts

TWEAK increases ITGA5 and ITGB1 transcripts in CD117+ cells, but surface alpha5beta1 heterodimer does not change. Give the mechanistic explanation the data support and the one they rule out.

REFERENCE ANSWER

The supported explanation is a surface ceiling: nearly all cells, CD117+ and CD117-, are already alpha5beta1-positive at baseline, so a flow readout scored as percent-positive cannot register an increase. Transcript can rise without changing the fraction of positive cells. What this rules out is the inference that TWEAK fails to regulate alpha5beta1 biology at all; the assay simply lacks dynamic range. The correct readout is surface density (median fluorescence intensity) or total protein rather than percent positive, and the paper's own use of the JBS-5 active-conformation antibody points at the further possibility that activation state, not abundance, is the regulated variable.

SCORING RUBRIC (20 pts)

- **[+5]** surface ceiling: nearly all cells already alpha5beta1+ at baseline
- **[+5]** percent-positive readout has no dynamic range to detect an increase
- **[+5]** does NOT rule out TWEAK regulation of alpha5beta1 biology
- **[+5]** correct readout is MFI/surface density or activation state, not percent positive

Lujano-Olazaba et al., Cell Reports 2026 · Figure 4F, Figure 4H

relb-M13 HARD · 25 pts

Beta1-active integrin is elevated on CD117+ cells but is independent of TWEAK, RelA, and RelB. What does this triple-independence imply, and why is it a load-bearing negative result?

REFERENCE ANSWER

Triple-independence means beta1 activation on CSCs is driven by something outside the TWEAK-Fn14-NF-kB axis entirely, and since ITGA5 changes do not explain it either, the likely partners are non-alpha5 alpha subunits paired with beta1. It is load-bearing because it establishes specificity: without it, one could argue TWEAK-Fn14-RelB simply raises integrin signaling in general. The negative result confines the axis to alphaVbeta3 and thereby makes the therapeutic argument for combining RelB blockade with alphaVbeta3 inhibition rather than broad integrin blockade. It also flags beta1 as a pre-existing, NF-kB-independent resistance channel that neither arm of the proposed combination touches.

SCORING RUBRIC (25 pts)

- **[+7]** beta1 activation driven outside the TWEAK-Fn14-NF-kB axis
- **[+6]** implicates non-alpha5 alpha partners for beta1
- **[+6]** establishes specificity: axis is not a general integrin amplifier
- **[+6]** flags beta1 as an untouched, NF-kB-independent resistance channel

Lujano-Olazaba et al., Cell Reports 2026 · Supplemental Figure 5C

relb-M14 MEDIUM · 20 pts

Loss of either RelA or RelB reduces alphaVbeta3 on CD117+ cells, but TWEAK-dependent enrichment preferentially requires RelB. Distinguish the two claims mechanistically.

REFERENCE ANSWER

These are claims about two different quantities. Baseline alphaVbeta3 depends on both arms, consistent with RelA supplying alphaV and RelB supplying beta3, so removing either subunit's driver lowers the heterodimer.

TWEAK-dependent enrichment is the increment above baseline in response to a specific ligand, and that increment requires RelB because TWEAK-Fn14 signals through the alternative arm. The distinction matters therapeutically: RelA blockade would lower the setpoint, while RelB blockade specifically severs the inducible, chemotherapy-triggered escalation. Since relapse is driven by post-chemotherapy TWEAK enrichment, severing the increment is the more targeted intervention, which is the paper's rationale for focusing on RelB.

SCORING RUBRIC (20 pts)

- **[+6]** baseline heterodimer requires both arms (alphaV from RelA, beta3 from RelB)
- **[+5]** TWEAK-induced increment specifically requires RelB via alternative signaling
- **[+4]** RelA blockade lowers setpoint; RelB blockade severs inducible escalation
- **[+5]** relapse is driven by the inducible increment -> RelB is the targeted choice

Lujano-Olazaba et al., Cell Reports 2026 · Figure 4G

relb-M15 HARD · 25 pts

RelB knockdown with chemotherapy increased omental tumor burden relative to chemotherapy alone, but adding Cilengitide abrogated that increase. Propose a mechanism that explains both halves.

REFERENCE ANSWER

A redistribution mechanism fits both. RelB knockdown suppresses mesenteric colonization, the RelB-dominant site, but does not kill the cells; they relocate to the omentum, where RelA dominates and alphaV-containing integrins other than alphaVbeta3 (notably alphaVbeta5, which is not RelB-dependent) can support colonization. Omental burden therefore rises. Adding Cilengitide blocks alphaVbeta3 and alphaVbeta5 together, closing the alphaV-dependent escape route at the omentum, so the increase disappears. The model predicts that RelB knockdown alone redistributes rather than eradicates, which is exactly why it failed to improve survival alone, and it makes the RelB-plus-Cilengitide synergy a case of blocking a primary site and its escape route simultaneously.

SCORING RUBRIC (25 pts)

- **[+7]** redistribution, not eradication: cells relocate from mesentery to omentum
- **[+6]** omentum is RelA-dominant and supports alphaVbeta5, which is not RelB-dependent
- **[+6]** Cilengitide blocks alphaVbeta3 AND alphaVbeta5, closing the escape route
- **[+6]** explains why shRelB alone failed on survival but synergized with Cilengitide

Lujano-Olazaba et al., Cell Reports 2026 · Figure 6G, Figure 7C

relb-M16 MEDIUM · 20 pts

Why is vitronectin enrichment in post-chemotherapy mesentery mechanistically more interesting than the COL1A1 enrichment, given this paper's model?

REFERENCE ANSWER

Vitronectin is a canonical RGD ligand for alphaV integrins, including both alphaVbeta3 and alphaVbeta5, the two receptors this paper implicates in stress tolerance. Its enrichment therefore closes the loop directly: chemotherapy remodels mesenteric matrix to deposit the specific ligand for the specific receptor the tumor cells are upregulating, at the specific site where RelB is dominant and regrowth is favored. COL1A1 signals matrix stiffening and is largely a beta1-integrin ligand, so it is relevant to the environment but not to the

alphaV axis under test. Vitronectin makes ligand and receptor rise together at the same site, which is the tightest available mechanistic correspondence in the dataset.

SCORING RUBRIC (20 pts)

- **[+5]** vitronectin is a canonical RGD ligand for alphaVbeta3 and alphaVbeta5
- **[+5]** its enrichment matches receptor upregulation at the same site (mesentery)
- **[+5]** COL1A1 is chiefly a beta1 ligand -> relevant to stiffening, not the alphaV axis
- **[+5]** closes the ligand-receptor loop: chemo deposits the ligand for the upregulated receptor

Lujano-Olazaba et al., Cell Reports 2026 · Figure 7I

relb-M17 MEDIUM · 20 pts

The paper argues alphaVbeta3 mediates isolation stress tolerance. Explain how the 1,500-cell inoculum design specifically tests isolation stress rather than proliferative capacity.

REFERENCE ANSWER

At 1,500 cells injected intraperitoneally, cells are dispersed, largely unanchored, and lack the paracrine and matrix support that a bulk inoculum self-supplies. The rate-limiting hurdle is surviving anchorage-independence and colonizing a niche, not dividing quickly. Two readouts separate these: stem cell frequency by ELDA scores whether a cell can start a tumor at all, which is a survival-and-colonization measure, while growth rate scores proliferation once established. AlphaVbeta3+ cells win on both ($1.39\text{e-}3$ vs $3.13\text{e-}4$ frequency; 268.5 vs 35.8 counts/day), but the frequency difference is the isolation-stress readout. A bulk inoculum would have masked the survival hurdle entirely.

SCORING RUBRIC (20 pts)

- **[+5]** low dispersed inoculum makes anchorage-independent survival rate-limiting, not division
- **[+5]** ELDA frequency measures survival/colonization; growth rate measures proliferation
- **[+5]** alphaVbeta3+ superior on both ($1.39\text{e-}3$ vs $3.13\text{e-}4$; 268.5 vs 35.8 counts/day)
- **[+5]** bulk inoculum would mask the survival hurdle

Lujano-Olazaba et al., Cell Reports 2026 · Figure 5E-H

relb-M18 HARD · 25 pts

Chemotherapy shifts CD117+ CSCs from alphaVbeta3+alphaVbeta5- to alphaVbeta3+alphaVbeta5+ double-positive. Given that alphaVbeta5 is reported to limit alphaVbeta3-mediated migration on vitronectin, what mechanistic model reconciles co-expression with that antagonism?

REFERENCE ANSWER

The antagonism and the co-expression are reconciled if the two receptors serve opposite phases of relapse. AlphaVbeta5 restraining alphaVbeta3-driven migration would favor a sessile, colonizing state rather than a disseminating one, which is what a cell needs immediately after chemotherapy: stay put, survive, and rebuild locally on remodeled matrix. Co-expression would then be a brake deliberately applied to a migratory receptor while retaining its survival signaling. The model predicts double-positive cells are less migratory but more stress-tolerant than alphaVbeta3-single-positive cells, and that alphaVbeta5 loss would restore migration while reducing local colonization; both are directly testable and would falsify the model if reversed.

SCORING RUBRIC (25 pts)

- **[+5]** receptors serve opposite phases: survival/colonization vs migration

- **[+8]** alphaVbeta5 acts as a brake on alphaVbeta3-driven migration -> sessile colonizing state
- **[+5]** post-chemotherapy the cell needs local rebuilding, not dissemination
- **[+7]** testable prediction: double-positives less migratory, more stress-tolerant

Lujano-Olazaba et al., Cell Reports 2026 · Figure 3G

relb-M19 MEDIUM · 20 pts

Why does the paper use the LM609 antibody rather than separate anti-alphaV and anti-beta3 antibodies, and what would be lost with the latter?

REFERENCE ANSWER

LM609 recognizes the assembled alphaVbeta3 heterodimer specifically, not either free subunit. Staining alphaV and beta3 separately would score subunit abundance, and since alphaV pairs promiscuously with beta1, beta5, beta6 and beta8, an alphaV-high cell need not carry any alphaVbeta3; double-positivity by two separate stains would not establish heterodimerization. Given that this paper's central claim is that two NF-kB arms each supply one subunit and the functional unit is the assembled receptor, the heterodimer-specific reagent is the only one that measures the quantity the model is about. LM609 also acts by steric hindrance of ligand binding, tying the marker to the function.

SCORING RUBRIC (20 pts)

- **[+4]** LM609 recognizes the assembled heterodimer, not free subunits
- **[+6]** alphaV pairs promiscuously -> alphaV-high does not imply alphaVbeta3
- **[+6]** separate stains cannot establish heterodimerization
- **[+4]** the model's central quantity is the assembled receptor, so the reagent must be dimer-specific

Lujano-Olazaba et al., Cell Reports 2026 · Figure 2C; Key Resources

relb-M20 HARD · 25 pts

TWEAK upregulates alpha5 protein in an NF-kB-dependent manner but downregulates beta4 independently of NF-kB. What does the NF-kB-independent arm of TWEAK signaling imply for the therapeutic model?

REFERENCE ANSWER

It establishes that Fn14 has outputs that RelA/RelB blockade will not touch, so blocking NF-kB is not equivalent to blocking TWEAK. Beta4 downregulation is a plausible relapse-relevant event on its own, since alpha6beta4 anchors cells to laminin in hemidesmosomes and its loss would license detachment and anchorage-independence, exactly the phenotype under study. The therapeutic implication is that RelB blockade leaves an NF-kB-independent Fn14 program intact, which is one reason RelB knockdown alone did not improve survival, and it argues that targeting Fn14 or TWEAK upstream would be mechanistically broader than targeting RelB, at the cost of specificity.

SCORING RUBRIC (25 pts)

- **[+7]** Fn14 has NF-kB-independent outputs -> NF-kB blockade is not TWEAK blockade
- **[+6]** beta4 loss (alpha6beta4/hemidesmosome) licenses detachment and anchorage-independence
- **[+6]** RelB blockade leaves an NF-kB-independent Fn14 program intact
- **[+6]** upstream Fn14/TWEAK targeting would be broader but less specific

Lujano-Olazaba et al., Cell Reports 2026 · Supplemental Figure 5A

relb-M21 MEDIUM · 20 pts

The relapse model injects only 150,000-300,000 cells and starts chemotherapy at day 7. Explain the mechanistic reasoning behind both choices and what a high-burden model would have measured instead.

REFERENCE ANSWER

The low inoculum creates a chemosensitive window in which chemotherapy can drive genuine regression to minimal residual disease rather than mere stasis, and it imposes the isolation stress that models the post-debulking state. The day-7 delay ensures tumors are established above a 1e5 luminescence threshold, below which pilot tumors spontaneously regressed, so any later loss of signal is attributable to treatment rather than failed engraftment. A high-burden model would have measured tumor stasis and progression, not regression and relapse; without a true nadir there is no minimal residual disease phase and therefore nothing that corresponds to the clinical question of what survives chemotherapy and regrows.

SCORING RUBRIC (20 pts)

- **[+5]** low inoculum creates a chemosensitive window permitting true regression, not stasis
- **[+5]** low inoculum also imposes post-debulking isolation stress
- **[+5]** day-7 delay puts tumors above the 1e5 threshold below which they spontaneously regress
- **[+5]** high-burden models measure progression/stasis, not regression and relapse

Lujano-Olazaba et al., Cell Reports 2026 · Figure 3A, Figure 6A

relb-M22 HARD · 25 pts

Cilengitide has been reported to polarize macrophages toward an immunosuppressive M2 phenotype. How does this complicate the interpretation of the RelB-knockdown-plus-Cilengitide survival benefit, and does it complicate it symmetrically for the RelA arm?

REFERENCE ANSWER

It means Cilengitide has a tumor-extrinsic immune effect that could act in either direction, so the survival benefit cannot be attributed purely to alphaVbeta3/alphaVbeta5 blockade on tumor cells. Notably the complication is asymmetric. For the RelA arm it plausibly explains a paradox: Cilengitide reduced rather than enhanced the shRelA benefit, and RelA+alphaVbeta3+ cells rose at the omentum with Cilengitide, which M2-mediated tumor support could account for. For the RelB arm it works against the observed result, since an immunosuppressive drug effect should blunt, not amplify, the benefit, which makes the RelB synergy more impressive rather than less. Nude mice retain macrophages, so the mechanism is live in this model.

SCORING RUBRIC (25 pts)

- **[+7]** Cilengitide has tumor-extrinsic immune effects -> benefit not purely receptor blockade
- **[+6]** asymmetric: plausibly explains the RelA-arm paradox (Cilengitide reduced shRelA benefit)
- **[+6]** for the RelB arm it works against the observed result, strengthening the synergy claim
- **[+6]** nude mice retain macrophages, so the mechanism is operative in this model

Lujano-Olazaba et al., Cell Reports 2026 · Figure 6B, Figure 7F; Discussion

relb-M23 MEDIUM · 20 pts

Explain why the paper's own model predicts that Cilengitide monotherapy should fail, and how that prediction was borne out.

REFERENCE ANSWER

AlphaVbeta3 is a stress-tolerance receptor, and stress tolerance only matters when there is stress. In an untreated tumor there is no cytotoxic stress, most cells are not depending on alphaVbeta3 for survival, and adhesion is handled by alpha5beta1, which Cilengitide only weakly touches. Blocking a receptor that is not currently load-bearing should therefore do nothing. This was borne out at every level: no viability effect in 2D or 3D across concentrations, no synergy with chemotherapy in vitro, and in vivo Cilengitide-alone mice reached takedown only about a week after untreated controls. The receptor becomes a target only once chemotherapy has created the stress that makes cells depend on it.

SCORING RUBRIC (20 pts)

- **[+5]** stress-tolerance receptors only matter under stress; untreated tumors are unstressed
- **[+5]** adhesion is alpha5beta1-dependent, which Cilengitide only weakly affects
- **[+5]** borne out: no in vitro viability effect, no in vitro synergy, ~1 week in vivo delay alone
- **[+5]** chemotherapy creates the dependency that makes the receptor targetable

Lujano-Olazaba et al., Cell Reports 2026 · Supplemental Figure 6B-C, Figure 6B-C

relb-M24 HARD · 25 pts

Both RelA and RelB knockdown reduce adhesion to fibronectin, but blocking alpha5beta1 also reduces it while blocking alphaVbeta3 does not. Trace the mechanistic chain that connects NF-kB to fibronectin adhesion, and identify where it breaks.

REFERENCE ANSWER

The chain would run NF-kB to integrin transcription to surface receptor to fibronectin adhesion. It holds at the ends: NF-kB knockdown reduces adhesion, and alpha5beta1 is the receptor doing the adhering. It breaks in the middle, because the NF-kB-regulated integrins the paper identifies are ITGAV and ITGB3, and alphaVbeta3 blockade has no adhesion effect, while surface alpha5beta1 is unchanged by RelA or RelB loss. So NF-kB is not reducing adhesion by lowering alpha5beta1 abundance. The remaining candidates are inside-out regulation of alpha5beta1 activation state, NF-kB control of other matrisome or adhesion genes flagged in the GO analysis, or an indirect effect via autocrine fibronectin. The paper does not resolve which.

SCORING RUBRIC (25 pts)

- **[+5]** chain holds at both ends but breaks in the middle
- **[+6]** NF-kB-regulated integrins here are ITGAV/ITGB3, not alpha5beta1
- **[+8]** surface alpha5beta1 unchanged by RelA/RelB loss -> not an abundance mechanism
- **[+6]** remaining candidates: inside-out activation, other matrisome genes, autocrine FN1; unresolved

Lujano-Olazaba et al., Cell Reports 2026 · Figure 5B-D, Figure 4H

relb-M25 MEDIUM · 20 pts

CD117+ cells show higher baseline ITGB3 than CD117- cells, and TWEAK raises ITGB3 only in CD117+ cells. What does the CSC-restricted inducibility imply about chromatin or pathway state?

REFERENCE ANSWER

Restricted inducibility means CD117- cells either lack the receiving apparatus or lack a permissive locus. Since the ITGB3 response is RelB-dependent and TWEAK-Fn14-driven, the candidates are lower Fn14 expression on CD117- cells, lower NIK/p52 processing capacity, or an ITGB3 locus that is closed in non-CSCs. The elevated baseline in CD117+ cells favors the chromatin explanation: the locus is already permissive there, which is

exactly what makes it further inducible. This predicts that CSC-restricted ITGB3 inducibility should track with accessibility at the ITGB3 locus and with RelB occupancy, and it makes CSC state, not TWEAK exposure, the gating variable for the whole axis.

SCORING RUBRIC (20 pts)

- **[+5]** CD117- cells lack receiving apparatus (Fn14/NIK) or a permissive locus
- **[+5]** elevated CD117+ baseline favors the open-chromatin explanation
- **[+6]** predicts ITGB3 accessibility and RelB occupancy track with CSC state
- **[+4]** CSC state, not TWEAK exposure, gates the axis

Lujano-Olazaba et al., Cell Reports 2026 · Figure 4E-F

relb-M26 MEDIUM · 20 pts

TWEAK increases EGFR and COL6A2 in CD117+ cells but not VEGFR2, PDGFR, or FN1. Why is the EGFR result mechanistically notable in the context of beta3 integrin biology?

REFERENCE ANSWER

Beta3 integrin has a well-documented relationship with EGFR-axis signaling: a beta3-KRAS-RalB complex has been shown to drive tumor stemness and resistance to EGFR inhibition, meaning beta3 upregulation can uncouple cells from EGFR dependence while sustaining downstream RAS signaling. TWEAK raising both ITGB3 and EGFR in CD117+ cells therefore assembles both components of a known stemness-plus-drug-tolerance module in the same cells at the same time. The COL6A2 increase adds a matrix-remodeling output from the tumor cells themselves, complementing the host-derived COL1A1/vitronectin changes. The negatives (VEGFR2, PDGFR, FN1) matter because they show TWEAK is not simply raising growth-factor receptors indiscriminately.

SCORING RUBRIC (20 pts)

- **[+5]** beta3-KRAS-RalB is an established stemness/EGFR-inhibitor-resistance axis
- **[+5]** TWEAK co-induces both ITGB3 and EGFR in the same CD117+ cells
- **[+5]** COL6A2 adds tumor-intrinsic matrix remodeling to host-derived COL1A1/VTN changes
- **[+5]** VEGFR2/PDGFR/FN1 negatives show the response is selective, not indiscriminate

Lujano-Olazaba et al., Cell Reports 2026 · Supplemental Figure 4G

relb-M27 HARD · 25 pts

The relapse model shows luminescence falling below background by day 25 and staying there ten days before regrowth. What does this plateau represent mechanistically, and why is it the most valuable window in the model?

REFERENCE ANSWER

The plateau is minimal residual disease: cells persist but are below the detection floor of bioluminescence, neither eradicated nor expanding. Mechanistically it is the state in which survivors are stress-tolerant and quiescent rather than proliferating, which is precisely the regime where RelB dominates over RelA in this system. It is the most valuable window because it is the only interval in which the cells that actually cause relapse are the only cells present; before it they are diluted by bulk tumor, after it they are diluted by regrown progeny. That dilution is exactly what the paper invokes to explain why CD117/CD133 enrichment was non-significant at the four-week harvest, and it argues the model's power is being spent at the wrong timepoint.

SCORING RUBRIC (25 pts)

- **[+6]** plateau = minimal residual disease below the bioluminescence detection floor
- **[+7]** survivors are quiescent and stress-tolerant -> the RelB-dominant regime
- **[+6]** the only window where relapse-causing cells are not diluted by bulk or progeny
- **[+6]** explains the non-significant CD117/CD133 result at the 4-week harvest

Lujano-Olazaba et al., Cell Reports 2026 · Figure 3B-C, Supplemental Figure 3B-C

relb-M28 MEDIUM · 20 pts

Why is p52 relevant to this paper's model even though the study measures RelB rather than p52, and what does the p52 prognostic literature add?

REFERENCE ANSWER

RelB does not act alone; the functional alternative NF-kB dimer is RelB:p52, generated by NIK-driven processing of p100. Measuring RelB abundance or nuclear localization without p52 leaves open whether the relevant variable is RelB protein or the availability of its obligate partner, and p100 processing is the actual rate-limiting, ligand-regulated step downstream of Fn14. The prognostic literature adds independent clinical weight: p52 is associated with worse overall and progression-free survival in HGSOC independently of RelA, which corroborates the paper's argument that the alternative arm is not merely a downstream echo of the classical arm but carries its own outcome-relevant signal.

SCORING RUBRIC (20 pts)

- **[+6]** the functional dimer is RelB:p52; p100->p52 processing is the ligand-regulated rate-limiting step
- **[+5]** RelB measured alone cannot distinguish RelB abundance from partner availability
- **[+5]** p52 predicts worse OS/PFS in HGSOC independently of RelA
- **[+4]** supports the alternative arm carrying independent, outcome-relevant signal

Lujano-Olazaba et al., Cell Reports 2026 · Discussion

relb-M29 HARD · 25 pts

The model claims chemotherapy shifts tumor growth from omentum to mesentery. Given that omentectomy is standard of care in HGSOC debulking, what does this add clinically that the mouse anatomy alone would not?

REFERENCE ANSWER

In patients the omentum is typically removed at debulking, so the site the mouse model shows to be chemo-suppressed is often surgically absent anyway; the clinically live question is what happens in the tissue left behind. The mesentery is not resected, and this model says it is exactly where chemotherapy-tolerant, RelB-high, alphaVbeta3+ cells preferentially regrow, and where chemotherapy remodels matrix (COL1A1, vitronectin) into a more protective substrate. The finding therefore converts an anatomical curiosity into a targeting rationale: standard care removes the favorable site and leaves the dangerous one, and the residual mesenteric niche is both the likely source of relapse and a rationale for improved intraoperative detection of mesenteric micrometastases.

SCORING RUBRIC (25 pts)

- **[+7]** omentum is usually resected at debulking; the mesentery is not
- **[+6]** the model localizes chemo-tolerant RelB-high alphaVbeta3+ regrowth to the retained tissue
- **[+6]** chemotherapy also remodels mesenteric matrix into a protective substrate

- **[+6]** converts anatomy into a targeting rationale for the residual mesenteric niche

Lujano-Olazaba et al., Cell Reports 2026 · Figure 3H, Figure 7C, Figure 7I; Discussion

relb-M30 HARD · 25 pts

Construct the complete causal chain this paper proposes from chemotherapy administration to relapse at the mesentery, and name the single weakest link.

REFERENCE ANSWER

Chemotherapy kills bulk tumor and stresses host stroma. Myeloid cells release TWEAK; TWEAK engages Fn14 on surviving CD117+ CSCs; Fn14 activates alternative NF- κ B, mobilizing RelB (with p52); RelB transcribes ITGB3, which dimerizes with RelA-supplied α V to form α V β 3; α V β 3, with α V β 5 co-acquired, delivers outside-in survival signaling that confers isolation and cytotoxic stress tolerance; meanwhile chemotherapy remodels mesenteric matrix, enriching vitronectin and COL1A1 to supply α V ligands; RelB-high α V β 3+ cells preferentially colonize that primed mesenteric niche and regrow. The weakest link is the α V β 3-to-survival-signaling step: the paper shows α V β 3 marks and predicts stress tolerance but never demonstrates the downstream effector (no FAK, Src, or RalB readouts), so the receptor's causal mechanism remains inferred from prior literature rather than measured here.

SCORING RUBRIC (25 pts)

- **[+8]** chemo -> myeloid TWEAK -> Fn14 -> alternative NF- κ B/RelB -> ITGB3 -> α V β 3 with RelA-supplied α V
- **[+5]** parallel arm: chemo remodels mesenteric matrix (VTN, COL1A1) supplying α V ligands
- **[+5]** RelB-high α V β 3+ cells colonize the primed mesenteric niche and regrow
- **[+7]** weakest link: no downstream effector measured (FAK/Src/RalB) -> receptor mechanism is inferred, not shown

Lujano-Olazaba et al., Cell Reports 2026 · Figures 1-7

Data Interpretation

30 questions · relb-D01–relb-D30

relb-D01 MEDIUM · 20 pts

ELDA gave stem cell frequencies of 1.39×10^{-3} for $\alpha\text{V}\beta 3^+$ and 3.13×10^{-4} for $\alpha\text{V}\beta 3^-$ cells, from a single 1,500-cell dose with $n=8$, $\text{chisq } 4.56$, $\text{DF } 1$, $P=0.0328$. What can and cannot be concluded, and what is the design's real vulnerability?

REFERENCE ANSWER

A single dose is mathematically sufficient for frequency estimation provided both hits and fails occur, which they did (7/8 and 3/8), so the maximum likelihood estimate and its confidence intervals are legitimate and the ~4.4-fold difference is real at $P=0.0328$. What cannot be concluded is that the single-hit Poisson model actually holds, because testing linearity requires multiple doses; the frequency estimates are model-dependent extrapolations from one point. The real vulnerability is not the statistics but the interpretation: the assay conflates survival and proliferation, since the readout is reaching 5×10^4 luminescence counts, and $\alpha\text{V}\beta 3^+$ cells also grow 7.5-fold faster once established. A slow-growing cell that engrafted might simply not have crossed threshold by day 88.

SCORING RUBRIC (20 pts)

- **[+4]** single dose is valid for frequency estimation when both hits and fails occur (7/8 vs 3/8)
- **[+8]** cannot verify the single-hit Poisson model without multiple doses -> model-dependent
- **[+4]** readout conflates survival with proliferation (threshold-based, and growth rates differ 7.5-fold)
- **[+4]** slow engrafters may be scored as non-engrafters at day 88

Lujano-Olazaba et al., Cell Reports 2026 · Figure 5G-H

relb-D02 HARD · 25 pts

$\alpha\text{V}\beta 3^+$ cells comprise ~94% of relapsed tumors but are a minority pre-treatment. Give the competing interpretations and state precisely what experiment distinguishes them.

REFERENCE ANSWER

Selection holds that $\alpha\text{V}\beta 3^+$ cells pre-existed, resisted chemotherapy, and survived while $\alpha\text{V}\beta 3^-$ cells died. Induction/plasticity holds that chemotherapy-driven NF- κB activation upregulated $\alpha\text{V}\beta 3$ de novo on survivors. Endpoint frequency data cannot separate these because both produce the same final distribution. The distinguishing experiment is lineage tracing: barcode or fluorescently mark sorted $\alpha\text{V}\beta 3^-$ cells before treatment and ask whether their progeny appear as $\alpha\text{V}\beta 3^+$ at relapse. This paper's own data argue both operate: sorted $\alpha\text{V}\beta 3^-$ cells still formed tumors in 3/8 mice, which requires either plasticity or impure sorting, while chemotherapy raised $\alpha\text{V}\beta 3$ in vitro on a fixed population, which is induction by definition.

SCORING RUBRIC (25 pts)

- **[+6]** selection vs induction/plasticity produce identical endpoint distributions
- **[+7]** endpoint frequency data cannot distinguish them
- **[+6]** lineage tracing/barcoding of pre-treatment $\alpha\text{V}\beta 3^-$ cells is the distinguishing experiment
- **[+6]** paper's own data support both: 3/8 $\alpha\text{V}\beta 3^-$ tumors formed, and chemo induces $\alpha\text{V}\beta 3$ in vitro

Lujano-Olazaba et al., Cell Reports 2026 · Figure 3E, Figure 2E, Figure 5G

relb-D03 MEDIUM · 20 pts

Beta1-active integrin increased after chemotherapy in the relapse model but the change was not significant ($p=0.1269$), while alphaVbeta3 and alphaVbeta5 were significant. How should the beta1 result be reported, and what is the interpretive trap?

REFERENCE ANSWER

It should be reported as an increase that did not reach significance, with the n and p value stated, and it should not be described as no change. The trap runs both ways. Reading it as a negative result overclaims: $p=0.1269$ with $n=4-5$ is consistent with a real effect the study was underpowered to detect, and the paper elsewhere shows beta1-active is elevated on CD117+ cells. Reading it as a positive trend to support the model overclaims in the other direction. The correct use is as a specificity contrast, weaker than the significant alphaVbeta3/alphaVbeta5 results: the data are consistent with alphaV integrins responding more than beta1, but the study cannot exclude a beta1 effect.

SCORING RUBRIC (20 pts)

- **[+4]** report as non-significant increase with n and p, not as 'no change'
- **[+8]** $n=4-5$ with $p=0.1269$ cannot exclude a real effect -> underpowered, not negative
- **[+4]** must not be spun as a supporting trend either
- **[+4]** usable only as a weaker specificity contrast against the significant alphaV results

Lujano-Olazaba et al., Cell Reports 2026 · Figure 3E, Supplemental Figure 5C

relb-D04 HARD · 25 pts

CD117 and CD133 increased after chemotherapy but not significantly. The authors attribute this to harvest timing rather than absence of response. Evaluate whether that explanation is a post hoc rescue or a principled prediction.

REFERENCE ANSWER

It is principled rather than post hoc, but only conditionally. The claim rests on prior published work from the same group showing chemotherapy-induced CD133/CD117 enrichment is transient and detected within a week of completing chemotherapy, whereas cells here were harvested four weeks after the last dose, with regrowth already underway and expanding non-CSCs diluting the fraction. That is a pre-existing, independently established kinetic model, not an explanation invented to absorb this result. What keeps it from being fully principled is that it was not tested in this study: no interim timepoint was harvested. The explanation is therefore plausible and prior-supported but unverified here, and it should be stated as such rather than asserted.

SCORING RUBRIC (25 pts)

- **[+7]** rests on prior independent published kinetics, not an invented rescue
- **[+6]** 4-week harvest vs prior detection within 1 week; regrowth dilutes the CSC fraction
- **[+6]** not tested in this study: no interim timepoint harvested
- **[+6]** should be reported as prior-supported but unverified here

Lujano-Olazaba et al., Cell Reports 2026 · Supplemental Figure 3B-C

relb-D05 MEDIUM · 20 pts

TCGA analysis shows high ITGB3 associates with worse progression-free survival: HR 1.23, CI 1.04-1.45, logrank $P=0.014$, $N=1001$. How much weight should this carry in the paper's argument?

REFERENCE ANSWER

Modest weight, and only as corroboration. The hazard ratio of 1.23 is a small effect and the confidence interval's lower bound of 1.04 nearly touches unity, so the association is statistically detectable largely because N=1001 is large, not because the effect is strong. It is bulk tumor expression, so it cannot speak to the CSC-restricted, chemotherapy-induced, RelB-dependent mechanism the paper proposes; ITGB3-high bulk tumors need not be enriched for the rare stress-tolerant population under study. It also cannot separate causation from marker status. Its legitimate role is showing the gene is not clinically irrelevant, which is a floor, not support for the mechanism.

SCORING RUBRIC (20 pts)

- **[+5]** HR 1.23 is a weak effect; CI lower bound 1.04 nearly touches unity
- **[+4]** significance is driven by N=1001, not effect size
- **[+6]** bulk expression cannot speak to a rare CSC-restricted, chemo-induced mechanism
- **[+5]** legitimate role is a relevance floor, not mechanistic support

Lujano-Olazaba et al., Cell Reports 2026 · Figure 2B

relb-D06 HARD · 25 pts

Reviewer 2 objected that spheroid formation was reduced by only ~15% yet was described as 'lost.' The authors revised to 'reduced.' Explain why this is a substantive scientific correction rather than a wording fix.

REFERENCE ANSWER

It changes what the data can support. Loss implies necessity: NF-κB is required for spheroid formation, and cells without it cannot form spheroids. A 15% reduction implies modulation: NF-κB raises the probability of spheroid initiation while most spheroids form without it, so it is one contributing input among several. That distinction propagates. If NF-κB were necessary, RelB knockdown alone should have been therapeutically sufficient; because it is merely contributory, the requirement for combination with Cilengitide is predicted rather than surprising. The authors' own follow-up sharpens it further: spheroids that did form showed no size difference across groups, so NF-κB affects initiation frequency but not subsequent growth.

SCORING RUBRIC (25 pts)

- **[+5]** 'lost' implies necessity; '~15% reduced' implies probabilistic modulation
- **[+8]** the distinction propagates to therapeutic prediction (single-agent sufficiency)
- **[+5]** explains why RelB knockdown alone was insufficient without Cilengitide
- **[+7]** spheroid size unchanged -> NF-κB affects initiation frequency, not growth

Lujano-Olazaba et al., Cell Reports 2026 · Supplemental Figure 2C-D

relb-D07 MEDIUM · 20 pts

In OV90, alphaVbeta3 enrichment in 3D is reduced by both shRelA and shRelB; in OVCAR8, only shRelB reduces it. How should this cell-line discordance be handled?

REFERENCE ANSWER

It should be reported as discordance and used to grade confidence, not averaged away or resolved by picking the convenient line. The RelB-to-ITGB3 relationship replicates across both lines and is therefore the robust claim; the RelA contribution is line-dependent and therefore the weaker one. That asymmetry is one of the paper's stated reasons for focusing on RelB, and it is an honest use of the discordance. The interpretive trap

is treating OV90 as showing a real RelA effect that OVCAR8 was underpowered to detect, versus treating OVCAR8 as showing RelA is dispensable; both are single-line inferences. The defensible statement is that RelB is consistently required while RelA's contribution varies by genetic background.

SCORING RUBRIC (20 pts)

- **[+5]** report the discordance; use it to grade confidence rather than average or cherry-pick
- **[+6]** RelB->ITGB3 replicates across lines = robust; RelA contribution is line-dependent = weaker
- **[+4]** this asymmetry is an explicit reason the paper prioritizes RelB
- **[+5]** defensible claim: RelB consistently required, RelA context-dependent

Lujano-Olazaba et al., Cell Reports 2026 · Figure 2C, Supplemental Figure 2B

relb-D08 HARD · 25 pts

Chemosensitivity was 93% loss for shRelA+Cilengitide and 89% for shRelB+Cilengitide, yet shRelB+Cilengitide gave far better survival (104 vs 57 days). Explain why the better chemosensitizer is not the better treatment.

REFERENCE ANSWER

The two endpoints measure different things at different times. Chemosensitivity is the depth of the nadir, a snapshot of how many cells chemotherapy kills. Survival is governed by what the survivors do afterward: whether they persist, where they colonize, and how fast they regrow. A 4-point difference in nadir depth is trivial compared with the biology of the residual population. The result is in fact the paper's most important internal finding, because it demonstrates that killing more cells is not the mechanism of benefit; the RelB+Cilengitide combination wins by disabling the survivors' regrowth capacity, specifically alphaVbeta3+ RelB+ colonization of the mesentery, not by deepening the kill. Any trial powered on response depth would miss this drug.

SCORING RUBRIC (25 pts)

- **[+6]** chemosensitivity measures nadir depth; survival measures survivor behavior afterward
- **[+6]** a 4-point nadir difference is trivial versus residual-population biology
- **[+7]** benefit mechanism is disabling regrowth (mesenteric alphaVbeta3+RelB+ colonization), not deeper kill
- **[+6]** a trial powered on response depth would miss this drug

Lujano-Olazaba et al., Cell Reports 2026 · Figure 6B-D, Figure 7C

relb-D09 MEDIUM · 20 pts

Reviewer 2 found that stated claims about ITGA5 and about 'relatively high' ITGB1/ITGB5/ITGAV expression were not supported by the figures shown. What does the nature of these errors tell you about how to read cross-platform expression claims?

REFERENCE ANSWER

Both errors came from blending two datasets with different gene coverage: ITGA5 was claimed from a Nanostring panel that did not contain it, and expression-level claims were made for genes absent from the panel shown. The general lesson is that a targeted panel and an unbiased RNA-seq study are not interchangeable, and any cross-platform claim must specify which platform supports which gene. A targeted panel cannot support absolute expression ranking at all, since it only contains preselected genes; only the RNA-seq, which captures the full integrinome, can support ITGB1 being highest. The corrected text does exactly this, narrowing absolute-level claims to the RNA-seq and listing cross-validated genes separately.

SCORING RUBRIC (20 pts)

- **[+5]** errors arose from blending datasets with different gene coverage
- **[+5]** a targeted panel cannot support absolute expression ranking (preselected genes only)
- **[+5]** cross-platform claims must state which platform supports which gene
- **[+5]** only the full-integrinome RNA-seq can support ITGB1 as highest

Lujano-Olazaba et al., Cell Reports 2026 · Figure 2A, Supplemental Figure 1D

relb-D10 HARD · 25 pts

TWEAK correlates with ITGB3 at $R^2=0.264$, ITGAV at 0.211, ITGA5 at 0.205, and ITGB1 at 0.013 ($p=0.478$), $N=18$. Interpret the whole pattern, not just the top hit.

REFERENCE ANSWER

The absolute values are weak: the best correlation explains 26% of variance, so TWEAK is at most one of several inputs to ITGB3 and the data cannot support TWEAK as a dominant driver. What carries the argument is the pattern rather than the magnitude. Three RGD-binding integrin subunits correlate at similar modest levels while ITGB1 is flat and not significant, and that dissociation is not what non-specific covariation with tumor content or inflammation would produce, since ITGB1 would rise too. The pattern is therefore a specificity argument, not a strength argument. With $N=18$ the confidence intervals on these R^2 values are wide, so even the rank ordering of ITGB3 above ITGAV should not be over-read.

SCORING RUBRIC (25 pts)

- **[+6]** weak absolute values: best $R^2=0.264$ explains only ~26% of variance
- **[+6]** the argument rests on the pattern (three RGD subunits up, ITGB1 flat), not magnitude
- **[+6]** ITGB1 dissociation argues against non-specific covariation with tumor content/inflammation
- **[+7]** $N=18 \rightarrow$ wide CIs; rank ordering of ITGB3 over ITGAV should not be over-read

Lujano-Olazaba et al., Cell Reports 2026 · Figure 4A

relb-D11 MEDIUM · 20 pts

Five of eight mice in the shRelB+Cilengitide group still had minimal luminescence at day 104 when the group was taken down. How does this censoring affect the survival claim?

REFERENCE ANSWER

It makes the reported benefit an underestimate rather than an overestimate. Those five animals had not reached the takedown threshold; the group was ended by protocol, not by events. The true median survival for that arm is therefore unknown and greater than 104 days, and the Kaplan-Meier estimate is right-censored. This strengthens rather than weakens the conclusion, since the observed 104-day figure is a floor. The interpretive obligation is to report it as censored rather than as a completed median, and the design lesson is that the study was terminated before the most effective arm could be characterized, which means the effect size is measurable only as a lower bound.

SCORING RUBRIC (20 pts)

- **[+5]** group ended by protocol, not by events $\rightarrow$ right-censored
- **[+5]** true median for that arm is unknown and >104 days
- **[+5]** the benefit is underestimated, so the conclusion is strengthened
- **[+5]** must be reported as censored, not as a completed median

Lujano-Olazaba et al., Cell Reports 2026 · Figure 6B-C

relb-D12 HARD · 25 pts

AlphaVbeta3+ cells reached tumor threshold at 268.5 luminescence counts per day versus 35.8 for alphaVbeta3- cells, but this was computed only among mice that formed tumors. Why does that conditioning matter?

REFERENCE ANSWER

Conditioning on tumor formation makes the comparison one between 7 alphaVbeta3+ mice and only 3 alphaVbeta3- mice, and those 3 are a selected subset: they are the alphaVbeta3- cells that overcame a barrier most of their group could not. That selection could plausibly enrich for cells that upregulated alphaVbeta3 after injection, in which case the growth-rate comparison is not between the sorted populations at all but between alphaVbeta3+ cells and alphaVbeta3+ revertants. It could equally select for cells with a compensating alternative survival route, which might be intrinsically slower. Either way n=3 with a selection filter cannot bear a 7.5-fold quantitative claim, and the honest statement is directional, not quantitative.

SCORING RUBRIC (25 pts)

- **[+5]** conditioning yields n=7 vs n=3, and the 3 are a selected non-random subset
- **[+8]** survivors may be alphaVbeta3+ revertants -> comparison may not be between sorted populations
- **[+5]** or they may use a compensating, intrinsically slower route
- **[+7]** n=3 with a selection filter cannot support a quantitative 7.5-fold claim; report directionally

Lujano-Olazaba et al., Cell Reports 2026 · Figure 5G-H

relb-D13 MEDIUM · 20 pts

Reviewer 1 asked whether phenotypes were compared against doxycycline-treated controls. Why is this control non-negotiable, and how did the authors resolve it?

REFERENCE ANSWER

Doxycycline is not inert. It inhibits mitochondrial translation, has documented anti-metalloproteinase activity, and can affect proliferation and matrix biology, all of which sit directly on top of this paper's endpoints of ECM organization, integrin expression, and stress tolerance. If knockdown cells receive doxycycline and controls do not, every difference is confounded by the drug, and given that the endpoints are matrix-related, doxycycline is an unusually plausible confounder here rather than a pro forma one. The authors resolved it correctly: all shNeg controls received doxycycline, and they added flow cytometry of alphaVbeta3 in shNeg cells with and without doxycycline showing no drug effect on the primary readout.

SCORING RUBRIC (20 pts)

- **[+5]** doxycycline inhibits mitochondrial translation and has anti-MMP/matrix activity
- **[+6]** those effects sit directly on this paper's ECM and integrin endpoints -> plausible, not pro forma
- **[+4]** without dox-treated controls every difference is drug-confounded
- **[+5]** resolved: all shNeg controls received dox, plus a direct no-effect control on alphaVbeta3

Lujano-Olazaba et al., Cell Reports 2026 · Supplemental Figure 2A

relb-D14 HARD · 25 pts

The CD117+ MACS fractions were only 40-75% pure and were not depleted of CD133+ cells. How does impurity affect the direction and magnitude of the reported CD117+ vs CD117- differences?

REFERENCE ANSWER

Impurity biases toward the null. If the CD117+ fraction contains up to 60% CD117- cells, the measured difference between fractions is diluted, so the true CD117+ enrichment of ITGB3 and the true CSC-restricted TWEAK inducibility are both larger than reported. Effects that reached significance despite this dilution are therefore conservative, and the argument is strengthened rather than weakened. Lack of CD133 depletion is a separate issue about which population was captured, not about bias direction: the authors justify it on the grounds that OVCAR8 expresses low CD133, so depletion would not have meaningfully changed composition. The one thing impurity does undermine is any quantitative claim about the magnitude of the difference.

SCORING RUBRIC (25 pts)

- **[+7]** impurity dilutes the contrast -> biases toward the null
- **[+6]** significant effects despite 40-75% purity are conservative underestimates
- **[+6]** CD133 non-depletion concerns population identity, not bias direction; justified by low CD133 in OVCAR8
- **[+6]** quantitative magnitude claims are undermined; directional claims are strengthened

Lujano-Olazaba et al., Cell Reports 2026 · Figure 4E-F; Methods

relb-D15 MEDIUM · 20 pts

Whole-spheroid flow cytometry used a >30 um size threshold and excluded PI-infiltrated spheroids. What could each criterion do to the result, and in which direction?

REFERENCE ANSWER

The size threshold excludes small aggregates and single cells, calibrated against a simultaneous single-cell control. If knockdown reduces spheroid size, small real spheroids would be filtered out, which would inflate the apparent reduction in spheroid number. The authors close this: spheroids that formed showed no size difference across groups, so the threshold is not doing selective work. The viability criterion excludes only fully PI-infiltrated spheroids, meaning the alphaVbeta3 readout is scored on living, organized structures rather than debris; that raises specificity but could hide a real effect if knockdown kills spheroids outright rather than preventing their formation. Both criteria are defensible, and the size control is what makes the first one so.

SCORING RUBRIC (20 pts)

- **[+5]** size threshold would inflate apparent reduction if knockdown shrank spheroids
- **[+5]** authors close this: no size difference among spheroids that formed
- **[+5]** viability exclusion raises specificity but could mask outright killing versus failed formation
- **[+5]** both defensible; the size control is what validates the threshold

Lujano-Olazaba et al., Cell Reports 2026 · Figure 2D, Supplemental Figure 2C-D; Methods

relb-D16 HARD · 25 pts

At day 25 chemotherapy-treated mice had no luminescent signal above background. What does 'no signal above background' license you to conclude about tumor burden?

REFERENCE ANSWER

Only that burden fell below the assay's detection floor, which is not zero. Bioluminescence sensitivity in vivo is limited by luciferin delivery, tissue depth, absorption, and a fixed 0.5-second exposure; deeply seated or poorly perfused cells can be invisible at counts far above zero, and post-chemotherapy perfusion changes could themselves reduce luciferin access independent of cell number. The subsequent regrowth proves cells were present throughout. The correct statement is undetectable disease, and the fact that regrowth followed

makes this the model's most important feature rather than a weakness: it establishes minimal residual disease. Reading it as eradication followed by re-seeding would be a serious error.

SCORING RUBRIC (25 pts)

- **[+6]** below detection floor, not zero; the floor is set by luciferin delivery, depth, absorption, fixed exposure
- **[+6]** post-chemo perfusion changes could reduce signal independent of cell number
- **[+7]** subsequent regrowth proves cells persisted -> minimal residual disease, not eradication
- **[+6]** 'undetectable' not 'absent'; reading it as eradication plus re-seeding would be a serious error

Lujano-Olazaba et al., Cell Reports 2026 · Figure 3B-C

relb-D17 MEDIUM · 20 pts

Organ burden was reported as percent of total luminescence per mouse. What does this normalization gain and what does it cost?

REFERENCE ANSWER

It gains within-animal control, removing variation in total tumor burden, luciferin uptake, and injection efficiency between mice, which makes distribution comparable across animals. It costs the ability to say anything about absolute burden, because the measure is compositional: percentages must sum to 100, so any site's apparent rise can be produced entirely by another site's fall. The reported mesenteric increase after chemotherapy is therefore inseparable from the concurrent peritoneal wall decrease; the mesentery may have grown, or it may simply have shrunk less than everything else. Distinguishing these requires absolute counts per organ alongside the percentages, which the compositional data alone cannot supply.

SCORING RUBRIC (20 pts)

- **[+5]** gains within-animal control over total burden, luciferin uptake, injection variability
- **[+5]** costs absolute burden information; the data are compositional and sum to 100%
- **[+5]** mesenteric rise is inseparable from the concurrent peritoneal wall fall
- **[+5]** resolving it requires absolute per-organ counts alongside percentages

Lujano-Olazaba et al., Cell Reports 2026 · Figure 3H, Figure 6G

relb-D18 HARD · 25 pts

RelB knockdown alone did not significantly reduce RelB+alphaVbeta3+ cells at the mesentery, but RelB knockdown plus Cilengitide did. Why is this internally strange, and what explanations survive?

REFERENCE ANSWER

It is strange because Cilengitide is an integrin inhibitor with no known transcriptional activity toward RelB, so adding it should not improve a knockdown's efficiency at depleting RelB+ cells. Three explanations survive. First, selection: Cilengitide kills or blocks alphaVbeta3+ cells generally, and since RelB+ cells are disproportionately alphaVbeta3+, the combination removes them by a second, integrin-dependent route rather than by better knockdown. Second, feedback: integrin outside-in signaling may sustain NF-kB activity, so blocking the receptor lowers RelB activity and makes the knockdown effective, which would imply a receptor-to-transcription-factor feedback loop the paper does not test. Third, the shRelB-alone comparison may simply be underpowered. The first two are mechanistically distinct and distinguishable.

SCORING RUBRIC (25 pts)

- **[+7]** strange because Cilengitide has no known transcriptional activity toward RelB

- **[+6]** selection: Cilengitide removes alphaVbeta3+ cells, which are disproportionately RelB+
- **[+6]** feedback: integrin outside-in signaling may sustain NF-kB, so blockade lowers RelB activity
- **[+6]** or underpowered comparison; the first two are distinguishable and the feedback loop is untested

Lujano-Olazaba et al., Cell Reports 2026 · Figure 7G

relb-D19 MEDIUM · 20 pts

The percentage of alphaVbeta3+ cells at the omentum was unchanged across all treatment groups, while the mesentery showed significant reduction with RelB knockdown. What does the omental null result contribute?

REFERENCE ANSWER

It is the internal specificity control that makes the mesenteric result interpretable. If alphaVbeta3 had fallen at both sites, the effect would read as a systemic consequence of RelB knockdown on tumor cells wherever they sit. Because it falls only at the mesentery, the effect is site-restricted, which supports the model that RelB's control of alphaVbeta3 requires a mesentery-specific input, most plausibly TWEAK-rich myeloid cells. The null also corroborates the RelA/RelB site gradient from the same figure, since the omentum is RelA-dominant and would not be expected to depend on RelB. Without the omental null, the mesenteric result would be far weaker.

SCORING RUBRIC (20 pts)

- **[+5]** omental null is the internal specificity control
- **[+5]** site-restricted effect rules out a systemic consequence of RelB knockdown
- **[+5]** supports a mesentery-specific input requirement (TWEAK-rich myeloid cells)
- **[+5]** corroborates the RelA-omentum / RelB-mesentery gradient in the same figure

Lujano-Olazaba et al., Cell Reports 2026 · Figure 7C-E

relb-D20 HARD · 25 pts

The ex vivo mesentery experiment used ID8 murine cells on murine matrix, whereas the entire rest of the paper uses human OVCAR8/CAOV4/OV90 cells. Evaluate this switch.

REFERENCE ANSWER

The switch is defensible for the specific question and costly for integration. It is defensible because the experiment tests whether host matrix is altered by chemotherapy, and using species-matched cells on murine matrix avoids the xenogeneic ligand-receptor mismatch that human cells on mouse matrix would introduce, and ID8 permits a syngeneic-compatible readout. It is costly because ID8 has never been characterized in this paper for alphaVbeta3 expression, NF-kB status, or CD117 positivity, so nothing licenses the assumption that ID8 reads the primed matrix through the same receptor. The experiment therefore establishes that chemo-remodeled matrix is protective in general, but cannot establish that it is protective via alphaVbeta3, which is what the model needs.

SCORING RUBRIC (25 pts)

- **[+6]** defensible: species-matched cells on murine matrix avoid xenogeneic ligand-receptor mismatch
- **[+6]** costly: ID8 uncharacterized here for alphaVbeta3, NF-kB, CD117
- **[+6]** establishes matrix priming is protective in general
- **[+7]** cannot establish it is protective via alphaVbeta3, which is what the model requires

Lujano-Olazaba et al., Cell Reports 2026 · Figure 7H

relb-D21 MEDIUM · 20 pts

COL1A1 and vitronectin rose with chemotherapy in both omentum and mesentery, but reached significance only in the mesentery. Does this support site-specificity?

REFERENCE ANSWER

Weakly, and it is a classic error to treat it as strong support. A difference in significance between two tissues is not the same as a significant difference between them; the omental change may be real but smaller, or equally large but noisier, and human tissue samples vary widely. Establishing site-specificity requires a direct test of the tissue-by-treatment interaction, not two separate within-tissue comparisons. The paper's own framing is careful here, stating the enrichment occurred in both tissues and was statistically significant in the mesentery. The data are consistent with site-specificity and also with a shared response of differing magnitude, and the analysis as run cannot distinguish them.

SCORING RUBRIC (20 pts)

- **[+4]** a difference in significance is not a significant difference
- **[+6]** site-specificity requires a tissue-by-treatment interaction test, not two within-tissue tests
- **[+5]** omental change may be real but smaller, or equally large but noisier
- **[+5]** data are consistent with both site-specificity and a shared response of differing magnitude

Lujano-Olazaba et al., Cell Reports 2026 · Figure 7I

relb-D22 HARD · 25 pts

Chemotherapy alone, chemotherapy+Cilengitide, chemotherapy+shRelB, and chemotherapy+Cilengitide+shRelA all reached takedown at days 50-57, while chemotherapy+shRelA reached day 85. What does the shRelA-alone result do to the paper's narrative?

REFERENCE ANSWER

It complicates it considerably. RelA knockdown with chemotherapy alone (day 85) outperformed RelB knockdown with chemotherapy alone (days 50-57) and outperformed both Cilengitide combinations except shRelB+Cilengitide. The paper's headline foregrounds RelB, but on single-knockdown survival RelA is the better target, which the authors acknowledge as unsurprising since RelA drives survival pathways synergistic with chemotherapy. The genuinely odd finding is that adding Cilengitide to shRelA made things worse, dropping it from day 85 back to the 50-57 window, which implies antagonism. That antagonism, not the RelB synergy, is the least explained result in the study, and the M2-polarization hypothesis is offered but untested.

SCORING RUBRIC (25 pts)

- **[+7]** shRelA+chemo (d85) beat shRelB+chemo (d50-57) -> RelA is the better single knockdown
- **[+6]** the RelB headline understates RelA's standalone performance
- **[+6]** adding Cilengitide to shRelA was antagonistic, dropping d85 back to d50-57
- **[+6]** that antagonism is the least explained result; M2 polarization is proposed but untested

Lujano-Olazaba et al., Cell Reports 2026 · Figure 6B-C

relb-D23 MEDIUM · 20 pts

Groups were taken down at different times (day 15 for shNeg, day 104 for shRelB+Cilengitide) once each reached average 5e5 luminescent counts. What does this staggered design do to cross-group tissue comparisons?

REFERENCE ANSWER

It equalizes tumor burden at harvest and confounds it with time. Comparing alphaVbeta3 or RelB positivity across groups is then a comparison at matched burden but at wildly different intervals since treatment, so differences could reflect tumor state at a given size or elapsed time and drift. The design choice is defensible, because harvesting at matched burden is arguably more meaningful than at matched time for a relapse question, but it must be stated. It also directly explains the CD117/CD133 dilution issue: groups harvested late are further into regrowth. Any cross-group phenotype claim inherits this confound and cannot be cleanly attributed to treatment.

SCORING RUBRIC (20 pts)

- **[+5]** equalizes burden at harvest while confounding it with elapsed time since treatment
- **[+5]** cross-group phenotype differences could reflect burden-matched state or time drift
- **[+5]** defensible for a relapse question but must be stated
- **[+5]** directly explains the CD117/CD133 dilution issue in late-harvested groups

Lujano-Olazaba et al., Cell Reports 2026 · Figure 6B-C, Figure 7C-G

relb-D24 HARD · 25 pts

Chemotherapy+Cilengitide significantly increased ovarian luminescence relative to chemotherapy alone in shNeg, and this was abrogated by either RelA or RelB knockdown. Interpret this apparent adverse effect.

REFERENCE ANSWER

On its face an integrin inhibitor increased burden at a site, which is the opposite of the intended effect. Two readings compete. Compositionally, the measure is percent of total, so an ovarian rise could be arithmetic: Cilengitide suppressed other sites and the ovary's share grew without absolute growth. Biologically, Cilengitide may have redistributed cells toward a site whose colonization does not require alphaV integrins, or its M2-polarizing effect may have created a permissive ovarian niche. The fact that both knockdowns abrogated it argues the effect is NF-kB-dependent and therefore more likely biological than arithmetic. Either way it is an adverse signal for Cilengitide monotherapy and a caution that blocking one route can open another.

SCORING RUBRIC (25 pts)

- **[+5]** an integrin inhibitor increasing site burden is the opposite of the intended effect
- **[+5]** compositional reading: percent-of-total arithmetic, other sites suppressed
- **[+5]** biological reading: redistribution to an alphaV-independent site, or M2-permissive niche
- **[+10]** abrogation by both knockdowns argues NF-kB-dependence -> more likely biological; blocking one route can open another

Lujano-Olazaba et al., Cell Reports 2026 · Figure 6G

relb-D25 MEDIUM · 20 pts

The paper states alphaVbeta3+ cells 'constitute 94% of regrown tumors at relapse.' What is the precise denominator, and how does that constrain the claim?

REFERENCE ANSWER

The denominator is RFP+ cells recovered from resected tumors, not all cells in the tumor. RFP marks the injected OVCAR8 line, so the figure is the percentage of injected-lineage tumor cells that are alphaVbeta3+,

excluding host stroma, immune cells, and vasculature entirely. This is the right denominator for the claim, since the question is what tumor cells survived, and the authors verify RFP+ cells were a similar share of total tumor across treatments so the gating is not itself treatment-biased. The constraint is that the number says nothing about tumor composition overall, and it is specific to a single cell line in a xenograft, so it cannot be read as 94% of a patient's relapsed tumor.

SCORING RUBRIC (20 pts)

- **[+4]** denominator is RFP+ injected-lineage tumor cells, not all cells in the tumor
- **[+4]** excludes host stroma, immune, vasculature by design
- **[+6]** authors verify RFP+ share of total tumor is similar across treatments -> gating not treatment-biased
- **[+6]** says nothing about overall tumor composition; single line, xenograft -> not generalizable to patient tumors

Lujano-Olazaba et al., Cell Reports 2026 · Figure 3E, Supplemental Figure 3A

relb-D26 HARD · 25 pts

RNA-seq for the GO analysis was done in OV90 only, at n=3, with FDR<0.05 and log2FC<-0.5, yielding ~1,800-2,200 regulated genes per condition. Evaluate what this foundation can support.

REFERENCE ANSWER

Roughly 2,000 differentially expressed genes from a single cell line at n=3 is a hypothesis-generating dataset, and the paper uses it correctly as such, treating GO output as a lead that flags integrins rather than as a conclusion. Three constraints follow. A log2FC threshold of -0.5 is permissive, admitting sub-1.5-fold changes whose biological meaning is unclear. n=3 gives limited power, so the gene lists are unstable and GO enrichment inherits that instability. Single-cell-line data cannot establish generality, which is exactly why reviewers pushed for OVCAR8 and why the authors added it to Figures 1C and 1E. The load-bearing claims rest on the downstream flow, in vivo, and survival data, not on this analysis.

SCORING RUBRIC (25 pts)

- **[+5]** ~2,000 DEGs from one line at n=3 is hypothesis-generating, and the paper uses it that way
- **[+5]** log2FC<-0.5 is permissive; sub-1.5-fold changes of unclear biological meaning
- **[+5]** n=3 gives unstable gene lists; GO enrichment inherits the instability
- **[+10]** single line cannot establish generality -> reviewers required OVCAR8 addition; load-bearing claims rest elsewhere

Lujano-Olazaba et al., Cell Reports 2026 · Figure 1A-D

relb-D27 MEDIUM · 20 pts

In vivo flow analyzed a minimum of 250,000 RFP+ events, versus 50,000-100,000 for in vitro samples. Why the difference, and what does it protect against?

REFERENCE ANSWER

The higher in vivo threshold exists because the population of interest is rare within a rare fraction: CD117+ cells are under 10% of RFP+ cells, and the readout is alphaVbeta3 within that subfraction. At 250,000 RFP+ events the CD117+ subfraction still contains tens of thousands of cells, so percentage estimates carry acceptable Poisson error. It protects against the specific failure mode reviewers raised, in which sorting rare subfractions yields cell counts so low that a handful of events drives an apparently significant percentage. In

vitro populations are less heterogeneous and need fewer events for the same precision, so the asymmetry is principled rather than arbitrary.

SCORING RUBRIC (20 pts)

- **[+5]** the population of interest is rare within a rare fraction (CD117+ is <10% of RFP+)
- **[+5]** 250,000 RFP+ events keeps the CD117+ subfraction large enough for acceptable Poisson error
- **[+5]** protects against a handful of events driving an apparently significant percentage
- **[+5]** in vitro populations are less heterogeneous -> fewer events suffice; asymmetry is principled

Lujano-Olazaba et al., Cell Reports 2026 · Methods; Figure 3G

relb-D28 HARD · 25 pts

The senior author of the alphaVbeta3 work (Cheresh) holds a patent application on blocking alphaV-beta3 activation and founded a company developing bispecific integrin antibodies. How should this bear on reading the paper?

REFERENCE ANSWER

It should sharpen scrutiny of the interpretive choices, not the data. Properly disclosed conflicts do not invalidate results, and the mouse experiments here are large, blinded for IVIS reading, and internally controlled. Where it warrants attention is in framing: the Discussion's advocacy for bivalent antibodies targeting alpha5beta1 and alphaVbeta3 is a specific therapeutic direction aligned with the disclosed commercial interest, and it should be read as a proposal rather than a data-driven conclusion. Notably, the paper reports several results that cut against a commercial narrative, including Cilengitide's complete failure as monotherapy and its antagonism with shRelA, which is evidence of honest reporting rather than the reverse.

SCORING RUBRIC (25 pts)

- **[+6]** disclosed conflicts sharpen scrutiny of framing, not validity of data
- **[+5]** experiments are large, blinded for IVIS, and internally controlled
- **[+8]** the bivalent-antibody advocacy in the Discussion aligns with the disclosed interest -> read as proposal
- **[+6]** the paper reports commercially unfavorable results (Cilengitide monotherapy failure, shRelA antagonism), indicating honest reporting

Lujano-Olazaba et al., Cell Reports 2026 · Declaration of Interests; Discussion

relb-D29 MEDIUM · 20 pts

Nude (Nu/J) mice were used throughout the in vivo work. Which of the paper's conclusions survive this, and which are weakened?

REFERENCE ANSWER

Nude mice lack T cells but retain macrophages, NK cells, and B cells. Conclusions about tumor-cell-intrinsic biology survive intact: alphaVbeta3-dependent tumor initiation, ELDA frequencies, RelB-dependent integrin expression, and the site preference for regrowth do not require T cells. Conclusions involving myeloid biology are partially preserved, since macrophages are present, which matters because the TWEAK source is myeloid and because the Cilengitide M2-polarization concern remains live. What is weakened is anything about immune-mediated clearance or T-cell contribution to relapse, and this matters most for the translational proposals in the Discussion, since ADCC-based antibodies and CAR-T strategies cannot be evaluated in this host at all.

SCORING RUBRIC (20 pts)

- **[+5]** nude mice lack T cells but retain macrophages, NK, and B cells
- **[+5]** tumor-intrinsic conclusions survive: initiation, ELDA, RelB-integrin regulation, site preference
- **[+5]** myeloid-dependent claims partially preserved (TWEAK source, Cilengitide M2 concern remain live)
- **[+5]** weakened: immune clearance and T-cell contribution; ADCC and CAR-T proposals untestable in this host

Lujano-Olazaba et al., Cell Reports 2026 · Methods; Discussion

relb-D30 HARD · 25 pts

The stated limitation is that findings derive from cell lines and animal models. Rank the paper's major claims from most to least vulnerable to that limitation.

REFERENCE ANSWER

Least vulnerable is the patient-derived integrin expression change post-NACT, since it comes from two independent human cohorts (Nanostring N=31; RNA-seq N=18) and the TCGA ITGB3 survival association, none of which depend on cell lines. Intermediate is the RelA/RelB integrin subunit assignment, which is cell-line-derived but replicated across three lines and corroborated by the human TWEAK-ITGB3 correlation. More vulnerable is the RelB-mesentery site preference, which depends on murine peritoneal anatomy and myeloid composition that differ from humans, though a second relapse model in CAOV4 reproduced increased mesenteric growth. Most vulnerable is the therapeutic claim, which rests on a single cell line, a single inhibitor, shRNA rather than a drug, and an immunodeficient host, and has no patient-sample validation at all.

SCORING RUBRIC (25 pts)

- **[+7]** least vulnerable: human post-NACT integrin changes (two cohorts) and TCGA ITGB3 association
- **[+6]** intermediate: RelA/RelB subunit assignment - cell-line-based but replicated in 3 lines plus human TWEAK correlation
- **[+6]** more vulnerable: RelB-mesentery site preference depends on murine anatomy/myeloid composition (CAOV4 partially reproduces)
- **[+6]** most vulnerable: therapeutic claim - one line, one inhibitor, shRNA not a drug, immunodeficient host, no patient validation

Lujano-Olazaba et al., Cell Reports 2026 · Limitations of the Study

Hypothesis

15 questions · relb-H01–relb-H15

relb-H01 HARD · 25 pts

Design the single most informative follow-up to the RelB+Cilengitide survival result, and state what outcome would falsify the model.

REFERENCE ANSWER

Replace shRNA with a pharmacologic alternative-pathway inhibitor (a NIK inhibitor) and run the full factorial in the relapse model: chemotherapy alone, chemo+NIKi, chemo+Cilengitide, chemo+NIKi+Cilengitide, with the trial ended by events rather than protocol so the best arm's median is not censored. This tests whether the synergy is a druggable axis or an artifact of shRNA, and NIK blockade hits p100 processing, the actual ligand-regulated step. Falsification: if chemo+NIKi+Cilengitide fails to outperform chemo+Cilengitide, the RelB contribution is not the alternative-pathway transcriptional output and the shRelB result reflects an off-target or knockdown-specific effect. A second falsifier: if the benefit persists in mice whose mesenteric TWEAK is depleted, the proposed TWEAK-Fn14-RelB chain is not the mechanism.

SCORING RUBRIC (25 pts)

- **[+5]** swap shRNA for a druggable NIK inhibitor hitting p100 processing
- **[+5]** full factorial in the relapse model, ended by events not protocol
- **[+8]** falsifier 1: NIKi+Cilengitide fails to beat Cilengitide alone
- **[+7]** falsifier 2: benefit persists when mesenteric TWEAK is depleted

Lujano-Olazaba et al., Cell Reports 2026 · Figure 6B

relb-H02 HARD · 25 pts

The paper cannot distinguish selection from plasticity for the alphaVbeta3 enrichment. Design the experiment that settles it and state what each outcome would mean therapeutically.

REFERENCE ANSWER

Barcode OVCAR8 at clonal resolution, sort alphaVbeta3- and alphaVbeta3+ fractions, inject each separately in the relapse model, and sequence barcodes plus re-sort by alphaVbeta3 at relapse. If barcodes seeded as alphaVbeta3- reappear as alphaVbeta3+, plasticity operates; if relapsed alphaVbeta3+ barcodes are exclusively those seeded alphaVbeta3+, selection dominates. The therapeutic stakes differ sharply. Under pure selection, alphaVbeta3 is a stable marker and a debulking-time depletion of alphaVbeta3+ cells could be curative. Under plasticity, depleting alphaVbeta3+ cells is futile because survivors regenerate the phenotype, and the target must instead be the inducing signal (TWEAK-Fn14-RelB) rather than the induced receptor. The paper's model implicitly assumes both, since it targets signal and receptor together.

SCORING RUBRIC (25 pts)

- **[+5]** clonal barcoding plus pre/post alphaVbeta3 sorting in the relapse model
- **[+5]** plasticity if alphaVbeta3--seeded barcodes reappear as alphaVbeta3+
- **[+8]** selection -> depleting alphaVbeta3+ cells at debulking could be curative
- **[+7]** plasticity -> must target the inducing signal (TWEAK-Fn14-RelB), not the induced receptor

Lujano-Olazaba et al., Cell Reports 2026 · Figure 3E, Figure 5G

relb-H03 MEDIUM · 20 pts

The paper never measures the downstream effector of alphaVbeta3 signaling. Propose the experiment that closes this gap and justify the choice of readout.

REFERENCE ANSWER

Sort alphaVbeta3+ and alphaVbeta3- CD117+ cells from post-chemotherapy relapse tumors and profile phospho-FAK (Y397), phospho-Src, and RalB activity by pulldown, then repeat under Cilengitide to test ligand-dependence. FAK is the right primary readout for three reasons: the paper's own introduction names FAK as the outside-in survival effector, a co-author's prior work establishes that FAK activity sustains platinum resistance in ovarian cancer, and FAK inhibitors are clinically available, so a positive result is immediately actionable. RalB is the right secondary readout because the beta3-KRAS-RalB complex is the established stemness/drug-tolerance axis in the alphaVbeta3 literature. Without this, alphaVbeta3 remains a validated marker whose causal mechanism is imported rather than demonstrated.

SCORING RUBRIC (20 pts)

- **[+5]** sort alphaVbeta3+/- CD117+ cells from relapse tumors; profile pFAK Y397, pSrc, RalB activity
- **[+5]** repeat under Cilengitide to establish ligand-dependence
- **[+5]** FAK justified: named in the paper's own model, co-author precedent in platinum resistance, clinically druggable
- **[+5]** RalB justified by the established beta3-KRAS-RalB stemness/drug-tolerance axis

Lujano-Olazaba et al., Cell Reports 2026 · Introduction; Figure 5

relb-H04 HARD · 25 pts

Given that RelB knockdown redistributes tumor to the omentum while Cilengitide closes that route, propose a therapeutic sequencing hypothesis that exploits the site asymmetry rather than blocking both sites simultaneously.

REFERENCE ANSWER

Hypothesis: since standard care already removes the omentum, and RelB blockade appears to push cells toward it, RelB inhibition given before omentectomy could actively concentrate stress-tolerant cells into the tissue about to be resected, converting a redistribution liability into a surgical advantage. This inverts the paper's framing, in which omental redistribution is the problem Cilengitide must fix. Test it by giving RelB inhibition during neoadjuvant chemotherapy, then performing interval debulking with omentectomy, and quantifying whether mesenteric alphaVbeta3+ burden is lower than with chemotherapy alone. The prediction is falsified if redistribution requires a longer interval than neoadjuvant timing allows, or if cells redistribute to non-resected sites such as the ovary or peritoneal wall instead.

SCORING RUBRIC (25 pts)

- **[+6]** exploit redistribution: RelB blockade drives cells to the omentum, which is resected anyway
- **[+6]** inverts the paper's framing of omental redistribution as a liability
- **[+6]** test via RelB inhibition during NACT followed by interval debulking with omentectomy
- **[+7]** falsified if the interval is too short, or if cells redistribute to non-resected sites instead

Lujano-Olazaba et al., Cell Reports 2026 · Figure 6G, Figure 7C; Discussion

relb-H05 MEDIUM · 20 pts

The relapse model's most informative window (the 10-day minimal residual disease plateau) was never sampled. Design the experiment that exploits it.

REFERENCE ANSWER

Run the relapse model with staged harvests: at nadir (day 25-35, during the plateau), at first detectable regrowth, and at takedown. At each timepoint, sort RFP+ cells and profile alphaVbeta3, alphaVbeta5, CD117, CD133, and nuclear RelA/RelB by imaging flow. Predictions: CSC markers should be maximally enriched at nadir and decline as non-CSCs expand, which would convert the paper's post hoc explanation of the non-significant CD117/CD133 result into a tested one; nuclear RelB should peak at nadir if quiescent survivors are RelB-driven; and alphaVbeta3 should already be near-saturating at nadir rather than rising during regrowth, which would favor selection over late induction. The obstacle is that cells at nadir are below bioluminescence detection, so harvesting requires blind dissection or a more sensitive tracer.

SCORING RUBRIC (20 pts)

- **[+5]** staged harvests at nadir, first regrowth, and takedown, sorting RFP+ cells at each
- **[+4]** tests the CD117/CD133 dilution explanation rather than asserting it
- **[+6]** nuclear RelB predicted to peak at nadir if quiescent survivors are RelB-driven
- **[+5]** alphaVbeta3 saturating at nadir would favor selection over late induction; obstacle is sub-detection burden at nadir

Lujano-Olazaba et al., Cell Reports 2026 · Figure 3B-C, Supplemental Figure 3B-C

relb-H06 HARD · 25 pts

Propose an experiment to determine whether site-specific NF-kB arm usage is driven by the niche or is a heritable cell-intrinsic state.

REFERENCE ANSWER

Harvest RFP+ cells from relapsed mesentery (RelB-high) and from omentum (RelA-high), then re-inject each into naive mice and assess arm usage at both sites after establishment. If mesentery-derived cells become RelA-high when they colonize the omentum, the niche dictates arm usage. If they retain RelB-high status regardless of site, the state is heritable and cells home to the matching site. A cleaner orthogonal test: implant decellularized mesenteric versus omental matrix subcutaneously in the same mouse, seed both with the same cells, and read RelA/RelB. That controls for site-of-origin homing and asks only whether matrix composition is sufficient. Measuring TWEAK and TNF protein at each site provides the ligand-level correlate either way.

SCORING RUBRIC (25 pts)

- **[+6]** cross-transfer relapsed mesentery-derived and omentum-derived cells into naive mice
- **[+6]** niche-driven if arm usage flips with site; heritable if it persists regardless
- **[+7]** orthogonal test: implant decellularized mesenteric vs omental matrix in the same mouse, seed identically
- **[+6]** measure TWEAK and TNF protein at each site as the ligand-level correlate

Lujano-Olazaba et al., Cell Reports 2026 · Figure 7D-E

relb-H07 MEDIUM · 20 pts

The alphaVbeta5 upregulation is the paper's most under-explored finding. Propose the experiment that would establish whether it is functional or a bystander.

REFERENCE ANSWER

Knock down ITGB5 in OVCAR8, then run the relapse model with and without RelB knockdown, reading survival, site distribution, and alphaVbeta3 status. The competing predictions are sharp. If alphaVbeta5 is

functionally redundant with alphaVbeta3 for stress tolerance, ITGB5 loss should phenocopy part of the Cilengitide benefit and should synergize with RelB knockdown, which would explain why Cilengitide (blocking both) beat a beta3-only intervention. If alphaVbeta5 is a bystander marker co-induced by stress, ITGB5 loss should do nothing. A third possibility follows from the reported alphaVbeta5-restrains-alphaVbeta3-migration finding: ITGB5 loss could increase dissemination while reducing local colonization, which would make it a poor target despite being a good marker.

SCORING RUBRIC (20 pts)

- **[+4]** ITGB5 knockdown in the relapse model, with and without RelB knockdown
- **[+5]** redundant with alphaVbeta3 -> phenocopies part of Cilengitide benefit and synergizes with shRelB
- **[+5]** bystander -> ITGB5 loss does nothing
- **[+6]** third outcome: ITGB5 loss increases dissemination while reducing colonization -> good marker, poor target

Lujano-Olazaba et al., Cell Reports 2026 · Figure 3E, Figure 3G

relb-H08 HARD · 25 pts

Cilengitide antagonized the shRelA survival benefit (day 85 down to days 50-57). Propose a hypothesis for this and the experiment that tests it.

REFERENCE ANSWER

Hypothesis: the antagonism is immune-mediated rather than tumor-intrinsic. Cilengitide polarizes macrophages toward an immunosuppressive M2 phenotype, and nude mice retain macrophages, so Cilengitide could supply tumor-supportive myeloid signaling that offsets the loss of RelA-driven survival. The paper's own observation that RelA+alphaVbeta3+ cells rose at the omentum with Cilengitide fits. Test it by repeating the shRelA arms with macrophage depletion (clodronate or CSF1R blockade). If Cilengitide's antagonism disappears when macrophages are depleted, the mechanism is myeloid. If it persists, the antagonism is tumor-intrinsic, and the alternative hypothesis is compensatory signaling in which RelA loss plus alphaV blockade together trigger a rescue pathway neither triggers alone.

SCORING RUBRIC (25 pts)

- **[+7]** hypothesis: Cilengitide-driven M2 polarization supplies tumor-supportive myeloid signaling; nude mice retain macrophages
- **[+6]** supported by the observed rise in RelA+alphaVbeta3+ cells at the omentum with Cilengitide
- **[+6]** test by repeating shRelA arms with clodronate or CSF1R-blockade macrophage depletion
- **[+6]** antagonism disappearing = myeloid mechanism; persisting = tumor-intrinsic compensatory signaling

Lujano-Olazaba et al., Cell Reports 2026 · Figure 6B, Figure 7F; Discussion

relb-H09 MEDIUM · 20 pts

The paper proposes bivalent antibodies targeting alpha5beta1 and alphaVbeta3. Given this paper's own data, argue both for and against that specific pairing.

REFERENCE ANSWER

For: the two receptors carry non-overlapping functions this paper separated cleanly, alpha5beta1 for fibronectin adhesion and alphaVbeta3 for stress-tolerance signaling, so a bivalent agent would block both anchoring and survival rather than one or the other, and Cilengitide's partial alpha5beta1 activity may already contribute to its benefit. Against: this paper never showed alpha5beta1 blockade helps in vivo; its

only alpha5beta1 data are in vitro adhesion. Worse, alpha5beta1 sits at a surface ceiling on essentially all cells including CD117- cells, so it is not selective for the stress-tolerant population and blockade would hit bulk tumor and normal tissue alike. Nothing in this dataset argues alpha5beta1 rather than alphaVbeta5 is the right partner, and the alphaVbeta5 co-induction data arguably point the other way.

SCORING RUBRIC (20 pts)

- **[+5]** for: non-overlapping functions (alpha5beta1 adhesion, alphaVbeta3 survival); Cilengitide's partial alpha5beta1 activity may already contribute
- **[+4]** against: no in vivo alpha5beta1 data in this paper, only in vitro adhesion
- **[+6]** against: alpha5beta1 is at a surface ceiling on nearly all cells -> not selective for stress-tolerant cells
- **[+5]** this dataset's alphaVbeta5 co-induction arguably favors beta5 over alpha5beta1 as the partner

Lujano-Olazaba et al., Cell Reports 2026 · Figure 4H, Figure 5C-D; Discussion

relb-H10 HARD · 25 pts

The stated limitation is a lack of patient-sample validation. Design the minimal patient-sample study that would most strengthen the paper's central claim.

REFERENCE ANSWER

Use patient-matched pre- and post-NACT tumor tissue and score, by multiplex imaging on the same sections, alphaVbeta3 (LM609), CD117, and nuclear RelB in tumor cells, stratified by anatomical site where mesenteric or peritoneal samples are available. The single most valuable result would be co-localization: nuclear RelB and alphaVbeta3 in the same CD117+ cells, enriched post-NACT. That would validate the mechanism's core coincidence in human tissue, which no existing cohort data address, since the Nanostring and RNA-seq studies are bulk transcript. Adding TWEAK/Fn14 staining would test the upstream ligand step. Feasibility is the constraint: matched pre/post NACT tissue is scarce, mesenteric sampling is rarer still, and nuclear RelB quantification in FFPE is technically demanding.

SCORING RUBRIC (25 pts)

- **[+6]** patient-matched pre/post-NACT tissue with multiplex imaging: alphaVbeta3 (LM609), CD117, nuclear RelB
- **[+6]** key result is co-localization in the same CD117+ cells, enriched post-NACT
- **[+7]** validates the mechanism's core coincidence, which bulk transcript cohorts cannot address
- **[+6]** constraints: scarce matched tissue, rare mesenteric sampling, difficult nuclear RelB quantification in FFPE

Lujano-Olazaba et al., Cell Reports 2026 · Limitations of the Study

relb-H11 MEDIUM · 20 pts

Given that chemotherapy-remodeled mesenteric matrix protects tumor cells independent of tumor presence, propose a prophylactic strategy and its key risk.

REFERENCE ANSWER

Strategy: target the stromal remodeling rather than the tumor cell, by inhibiting the matrix response to chemotherapy so the protective niche never forms. Candidates are LOX or LOXL2 inhibition to prevent collagen crosslinking, or blocking vitronectin deposition, timed concurrently with chemotherapy since the remodeling is chemotherapy-driven and occurs in tumor-naive tissue. The key risk is that matrix remodeling after cytotoxic injury is a physiological wound-healing response, so inhibiting it systemically could impair normal repair, including surgical wound healing in patients who have just undergone debulking, and could

affect the peritoneum broadly. The therapeutic window depends on whether the tumor-protective and repair functions of the response are separable, which the ex vivo data cannot address.

SCORING RUBRIC (20 pts)

- **[+5]** target stromal remodeling so the protective niche never forms; LOX/LOXL2 or vitronectin deposition, timed with chemotherapy
- **[+4]** justified because remodeling is chemo-driven and occurs in tumor-naive tissue
- **[+5]** key risk: the response is physiological wound healing; inhibition could impair post-debulking surgical repair
- **[+6]** window depends on separability of tumor-protective from repair functions, which the ex vivo data cannot address

Lujano-Olazaba et al., Cell Reports 2026 · Figure 7H-I

relb-H12 HARD · 25 pts

Propose the experiment that would distinguish whether Cilengitide's synergy with RelB knockdown is due to alphaVbeta3 blockade, alphaVbeta5 blockade, or both.

REFERENCE ANSWER

Replace the pan-RGD inhibitor with selective tools and run the relapse model on a RelB-knockdown background: LM609 (alphaVbeta3-selective, and already validated in this paper as a steric ligand-binding blocker), P1F6 or an equivalent alphaVbeta5-selective antibody, both together, and Cilengitide as the comparator. If LM609 alone reproduces the Cilengitide synergy, beta3 blockade suffices and the model is confirmed as stated. If only the combination reproduces it, alphaVbeta5 is a required co-target, which would matter because alphaVbeta5 is not RelB-dependent and therefore represents a genuinely separate axis. If neither reproduces it, Cilengitide's benefit derives from its alpha5beta1 activity or from off-target effects such as macrophage polarization, which would substantially revise the mechanism.

SCORING RUBRIC (25 pts)

- **[+6]** swap pan-RGD Cilengitide for selective LM609 (alphaVbeta3) and P1F6-type (alphaVbeta5) antibodies on a shRelB background
- **[+5]** LM609 alone reproducing synergy confirms the model as stated
- **[+6]** only-combination result makes alphaVbeta5 a required, RelB-independent co-target
- **[+8]** neither reproducing it implicates alpha5beta1 activity or off-target macrophage effects -> major revision

Lujano-Olazaba et al., Cell Reports 2026 · Figure 6B; Key Resources

relb-H13 MEDIUM · 20 pts

The paper's therapeutic claim rests on shRNA, not a drug. What is the most tractable path to a RelB-targeting agent, and what is the principal obstacle?

REFERENCE ANSWER

The most tractable path is not to drug RelB directly but to drug the pathway step upstream: NIK (MAP3K14) inhibition blocks p100 processing to p52, which removes RelB's obligate partner and is the ligand-regulated rate-limiting step downstream of Fn14, and NIK inhibitors exist. Alternatives are Fn14-blocking antibodies, which are further upstream but would also sever the NF-kB-independent Fn14 outputs this paper documents, and RelB degraders, which are the most direct but least developed. The principal obstacle for any of these is that alternative NF-kB governs lymphoid organogenesis, B-cell survival, and osteoclast biology, so systemic

blockade risks immunosuppression and bone effects; intermittent dosing timed to chemotherapy cycles, as the mouse design already implies, is the natural mitigation.

SCORING RUBRIC (20 pts)

- **[+5]** most tractable: NIK (MAP3K14) inhibition blocks p100->p52, removing RelB's obligate partner
- **[+5]** alternatives: Fn14 antibodies (broader, also cuts NF-kB-independent outputs) or RelB degraders (direct, least developed)
- **[+5]** principal obstacle: alternative NF-kB governs lymphoid organogenesis, B-cell survival, osteoclast biology
- **[+5]** mitigation: intermittent dosing timed to chemotherapy cycles, as the mouse design implies

Lujano-Olazaba et al., Cell Reports 2026 · Figure 6; Discussion

relb-H14 HARD · 25 pts

Suppose a trial of chemotherapy plus an alphaVbeta3-blocking agent in HGSOc shows no PFS benefit. Using this paper's model, propose the three most likely reasons and how each would have been detectable preclinically.

REFERENCE ANSWER

First, monotherapy insufficiency: this paper shows integrin blockade only works with concurrent RelB suppression, so a trial without an alternative-pathway partner is testing an arm the model already predicts fails, and this was fully visible preclinically in the days 50-57 result for chemo+Cilengitide. Second, wrong endpoint: the benefit here appears as delayed regrowth, not deeper response, so a trial powered on response rate or short-interval PFS would miss it, which the chemosensitivity-versus-survival dissociation predicted. Third, compensation: alphaVbeta5, beta1-active integrins, and NF-kB-independent Fn14 outputs all remain available, and the paper documents each as untouched by beta3-directed intervention. All three were detectable preclinically, which is the broader point about why prior integrin trials disappointed.

SCORING RUBRIC (25 pts)

- **[+6]** monotherapy insufficiency: needs concurrent RelB suppression; visible in the d50-57 chemo+Cilengitide arm
- **[+7]** wrong endpoint: benefit is delayed regrowth, not deeper response; predicted by the chemosensitivity/survival dissociation
- **[+6]** compensation: alphaVbeta5, beta1-active, and NF-kB-independent Fn14 outputs remain available and documented as untouched
- **[+6]** all three were preclinically detectable, which explains prior integrin trial failures

Lujano-Olazaba et al., Cell Reports 2026 · Figure 6B-D; Discussion

relb-H15 HARD · 25 pts

Generate a hypothesis that this paper's data support but the authors did not propose, and state how you would test it.

REFERENCE ANSWER

Hypothesis: alphaVbeta3 and alphaVbeta5 compete for a limiting alphaV pool, making the two receptors mutually constraining rather than independent. Support: RelA supplies alphaV while RelB supplies beta3, so beta3 availability, not alphaV, is rate-limiting for alphaVbeta3, and any beta5 induction would sequester shared alphaV. This predicts a specific, testable consequence the paper's therapeutic model does not anticipate: RelB knockdown, by lowering beta3, should free alphaV and increase alphaVbeta5 rather than

merely leaving it unchanged, which would be a compensation mechanism explaining why shRelB alone failed on survival. Test by measuring surface alphaVbeta5 and total alphaV after RelB knockdown, with and without ITGB3 rescue; the hypothesis is falsified if alphaVbeta5 does not rise when beta3 falls, or if total alphaV is not limiting.

SCORING RUBRIC (25 pts)

- **[+4]** hypothesis: alphaVbeta3 and alphaVbeta5 compete for a limiting shared alphaV pool
- **[+6]** supported by RelA supplying alphaV and RelB supplying beta3 -> beta3 is rate-limiting for alphaVbeta3
- **[+8]** predicts RelB knockdown frees alphaV and raises alphaVbeta5 -> a compensation route explaining shRelB's solo failure
- **[+7]** test: surface alphaVbeta5 and total alphaV after shRelB, with and without ITGB3 rescue; falsified if beta5 does not rise or alphaV is not limiting

Lujano-Olazaba et al., Cell Reports 2026 · Figure 1D, Figure 3G, Figure 6B
